

Embedding Computable Infinitary Equivalence into P-Groups

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EMBEDDING COMPUTABLE INFINITARY EQUIVALENCE INTO P -GROUPS

A Dissertation

Submitted to the Graduate School
of the University of Notre Dame
in Partial Fulfillment of the Requirements
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by

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Abstract

by

Steven M. VanDenDriessche

We examine the relation between the uniformity of a collection of operators witnessing Turing computable embeddings, and the existence of an operator witnessing the universality of a class. The primary equivalence relation studied here is *computable infinitary Σ_α equivalence*. This project of exploiting uniformity of Turing computable embeddings to construct a limit embedding is carried out entirely in the context of countable reduced abelian p -groups. One may look at this program as either a project in the computable structure theory of abelian p -groups, or as a project in the construction of *limits* of sequences of uniform Turing computable operators.

In an attempt to explore the boundary between computable infinitary Σ_α equivalence and isomorphism, we show that for any computable α , certain classes of countable reduced abelian p -groups are universal for $\sim_\alpha^c$ under Turing computable embedding. Further, the operators witnessing these embeddings are extremely uniform.

Exploiting the uniformity of the embeddings, we produce operators which are, in some sense, limits of the embeddings witnessing the universality of the classes of countable reduced abelian p -groups. This is approached in three different ways: transfinite recursion on ordinal notation, Barwise-Kreisel Compactness, and hyperarithmetical saturation. Finally, we work in admissible set theory, and use Barwise Compactness and Σ_A -saturation to generalize selected results.

CONTENTS

ACKNOWLEDGMENTS	iii
CHAPTER 1: COMPUTABLE STRUCTURE THEORY AND TURING COM- PUTABLE EMBEDDINGS	1
1.1 Turing Computable Embeddings	3
1.2 Computable Infinitary Equivalence	8
CHAPTER 2: UNIVERSALITY OF ABELIAN p -GROUPS	9
2.1 Universality of ApG for $\sim_{2m}^c$	9
2.2 Universality of ApG for $\sim_\alpha^c$	13
CHAPTER 3: UNIFORMITY AND THE RECURSION THEOREM	23
3.1 Transfinite Induction and the Recursion Theorem	23
3.2 Operator Agreement	26
3.3 Applying Transfinite Induction on Ordinal Notation	33
CHAPTER 4: COMPACTNESS, SATURATION AND OPERATORS	37
4.1 Analytic Sets and Compactness	38
4.2 Applying Compactness	40
4.3 Hyperarithmetical Saturation	44
CHAPTER 5: ADMISSIBLE SETS AND OPERATORS	50
5.1 Compactness and Saturation in Admissible Set Theory	50
5.2 Σ_1 is Necessary for Gregory's Theorem	52
5.3 Applying Barwise Compactness	55
5.4 A Question on Σ_A -Saturation	58
APPENDIX A: ABELIAN p -GROUPS	60
APPENDIX B: ORDINAL NOTATION	64
APPENDIX C: KRIPKE-PLATEK SET THEORY	66
BIBLIOGRAPHY	67

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CHAPTER 1

COMPUTABLE STRUCTURE THEORY AND TURING COMPUTABLE EMBEDDINGS

Computable structure theory can be described as the study of the computational complexity of algebraic and combinatorial structures, functions between those structures, and relations on those structures. In particular, techniques from computability theory are applied to model theoretic structures in order to explore the boundaries of what is algorithmically computable in the study of the structures at hand. Following the usual conventions in computability, we let $(\varphi_e)_{e \in \omega}$ denote the standard enumeration of all partial computable functions. For the unary partial computable function with oracle X , we use φ^X . We say identify structures with their atomic diagrams and say that a structure is *computable* precisely when its atomic diagram is computable as a set of natural numbers. We let $\mathcal{A}_i$ denote the structure having universe and atomic diagram computed by φ_i . All structures considered will be countable unless stated otherwise.

The arithmetical hierarchy is formed by iterating the jump operator on the empty set, and taking projections and complements. The n th jump of the empty set is denoted $\emptyset^n$, or Δ_{n+1}^0 . If one wishes to code the information contained in all the arithmetical hierarchy into a single set, the infinite join of the sets suffices: $\emptyset^\omega = \bigoplus_{i \in \omega} \emptyset^i$. This process can be further iterated to obtain $\emptyset^\alpha$ for all *computable ordinals* α . The sets c.e. or co-c.e. in these $\emptyset^\alpha$ form the *hyperarithmetical hierarchy*. Technically, we must use Kleene's system of ordinal notations, denoted $\mathcal{O}$, in order to rigorously define the sets $\emptyset^\alpha$ and this hierarchy. See Appendix B for a quick overview of this

system.

When investigating the relation between computable structures and formulas in the language of the structure, a class of infinitary formulas rises to prominence. The *computable infinitary formulas* allow finite nesting of quantifiers, and only finitely many free variables, but allow conjunctions and disjunctions over c.e. (computably enumerable) sets of formulas. A hierarchy of formulas is defined inductively. The finitary quantifier-free formulas are called Δ_0^c formulas. A c.e. disjunction of existential quantifiers with formulas of lower complexity is Σ_α^c . A c.e. conjunction of universal quantifiers with formulas of lower complexity is Π_α^c . Given a computable language L , the collection of computable infinitary L -formulas is denoted $L_{\omega_1\omega}^c$. Similarly, $L_{\omega_1\omega}$ is the class of formulas with *countable* (not necessarily c.e.) disjunctions and conjunctions, that still only has finite nesting of quantifiers. The complexity of these sentences can be classified as Σ_α or Π_α in the obvious way.

A relation defined in a computable structure by a Σ_α^c formula will be Σ_α^0 . If the structure $\mathcal{A}$ is not computable, then a relation defined by a Σ_α^c formula is Σ_α^0 relative to the atomic diagram of the structure. In this case we say that the relation has complexity $\Sigma_\alpha^0(\mathcal{A})$. These preservations of complexity also hold for relations defined by Π_α^c formulas.

Following convention in computable structure theory, unless otherwise stated, all languages discussed are computable, and all structures will have universe a subset of ω . Further, all classes of structures are closed under isomorphism (modulo having universe a subset of ω). The classes that we consider are “nice” in the sense each is axiomatized by a single computable infinitary sentence. For a more thorough discussion on the assumed background required to read the present work, see [5], the standard book on the subject.

1.1 Turing Computable Embeddings

In many branches of mathematics, a collection of structures is classified with respect to some equivalence relation. Examples include algebraic structures with respect to isomorphism, and topological structures with respect to homotopy. Given classes of structures, K and K' with equivalence relations E and E' , we say that the classification of K' with respect to E' is *at least as difficult* as the classification of K up to E if from a complete classification for K' , we can *definably* extract a classification for K . This intuitive notion was formalized as the *Borel embedding* in [16], using descriptive set theory as a framework, and interpreting *definable* to mean Borel. In [10], the *computable embedding* was developed as a computability-theoretic analogue of the Borel embedding, where the definable reduction notion is based on enumeration reducibility. Also mentioned in [10], is the *Turing computable embedding*, which takes Turing reducibility as the notion of definable reducibility. These two computable embeddings have been further studied in [30], [11], and [15]. The Turing computable embedding is a central object of study in this thesis. It is relevant to note that all three of the above embeddings were originally studied only with isomorphism in mind, as we will investigate a different class of equivalence relations. In the context of Borel embeddings, equivalence relations other than isomorphism have recently been an area of much interest (for example in [38], [17], and [32]). The study of equivalence relations other than isomorphism under Turing computable embedding is very young, with the only other example being Wright's analysis of tree paths and computable isomorphism in [40].

Definition 1. *Let K and K' be classes of countable structures with equivalence relations E and E' , respectively. A Turing computable embedding of K with respect to E into K' with respect to E' is a Turing computable operator $\Phi = \varphi_e$ such that*

1. *for every $\mathcal{A} \in K$, there is a $\mathcal{B} \in K'$ such that $\Phi(\mathcal{A}) = \varphi_e^{D(\mathcal{A})} = \chi_{D(\mathcal{B})}$,*

2. for all $\mathcal{A}, \mathcal{B} \in K$, $\mathcal{A}E\mathcal{B}$ if and only if $\Phi(\mathcal{A})E'\Phi(\mathcal{B})$.

When such an operator exists, we write $(K, E) \leq_{tc} (K', E')$. If we have both $(K, E) \leq_{tc} (K', E')$ and $(K', E') \leq_{tc} (K, E)$, then we write $(K, E) \equiv_{tc} (K', E')$. If $(K, E) \leq_{tc} (K', E')$, but $(K', E') \not\leq_{tc} (K, E)$, then we write $(K, E) <_{tc} (K', E')$. This, like the usual notion of Turing reducibility, is a preorder, so modding out by $\equiv_{tc}$ yields a partial order of degrees. A powerful aspect of both the computable embedding and the Turing computable embedding is the ability to distinguish and analyze classes of finite structures. In the Borel setting, it is assumed that all structures have universe ω . Moreover, all classes with $\aleph_0$ isomorphism types are equivalent under the Borel embedding, while the Turing computable embedding can distinguish between these. For example, the classes of number fields and $\mathbb{Q}$ -vector spaces are not equivalent under Turing computable embeddings, despite the fact that each has $\aleph_0$ isomorphism types. The degree structure induced by Turing computable embedding with respect to isomorphism was studied in [30], which included transferring many results from [16] and [10] to the setting of Turing computable embeddings. For example, analysis of a theorem from [10] shows the following.

Theorem 1.1.1 (Knight, Miller, Vanden Boom, [30]). *Let PF , FLO , NF , FVS , VS and LO be the classes of finite prime fields, finite linear orders, number fields, finite dimensional $\mathbb{Q}$ -vector spaces, $\mathbb{Q}$ -vector spaces, and linear orders, respectively. Then,*

$$(PF, \cong) <_{tc} (FLO, \cong) \equiv_{tc} (NF, \cong) <_{tc} (FVS, \cong) <_{tc} (VS, \cong) <_{tc} (LO, \cong).$$

Similarly, checking the effectiveness of theorems from [16] shows that there is a maximum in this degree structure.

Theorem 1.1.2 (H. Friedman and L. Stanley, [16]). *Let UG be the class of undirected*

graphs, and FC_0 be the class of fields of characteristic 0. Then for any class K ,

$$(K, \cong) \leq_{tc} (UG, \cong) \equiv_{tc} (LO, \cong) \equiv_{tc} (FC_0, \cong).$$

There are several ways to say that a class K lies in the maximum degree of this partial ordering. In particular, K is said to be *universal* or *complete* for $\leq_{tc}$ with respect to isomorphism. Knight, Miller, and Vanden Boom use forcing in [30] to prove a theorem that is useful for showing that an embedding between classes cannot exist. For a highly readable exposition of a special case of the proof, see [39].

Theorem 1.1.3 (Pullback Theorem; Knight, Miller, Vanden Boom, [30]). *If $(K, E) \leq_{tc} (K', E')$ via $\Phi = \varphi_e$, then for any computable infinitary sentence φ in the language of K' , we can find a computable infinitary sentence φ^* in the language of K such that for all $\mathcal{A} \in K$, $\Phi(\mathcal{A}) \models \varphi$ if and only if $\mathcal{A} \models \varphi^*$. Further, if φ is Σ_α^c or Π_α^c , for $\alpha \geq 1$, then so is φ^* .*

The following corollary illustrates the use of the Pullback Theorem in showing that one class cannot Turing computably embed in another.

Corollary 1.1.4. *Let K and K' be classes of structures and α some fixed computable ordinal such that*

- (1) *for any $\mathcal{A}$ and $\mathcal{B}$ in K' that are not isomorphic, there exists a Σ_α^c sentence ψ such that $\mathcal{A} \models \psi$ and $\mathcal{B} \not\models \psi$,*
- (2) *there are structures, $\mathcal{C}$ and $\mathcal{D}$ in K that are not isomorphic, but satisfy the same Σ_α^c sentences.*

Then $(K, \cong) \not\leq_{tc} (K', \cong)$.

Proof. Suppose that such an embedding does exist, say $\Phi = \varphi_e$, and consider the pair of structures from condition (2). Since $\mathcal{C} \not\cong \mathcal{D}$, we have that $\Phi(\mathcal{C}) \not\cong \Phi(\mathcal{D})$, and thus by (1) that there is some Σ_α^c sentence ψ such that $\Phi(\mathcal{C}) \models \psi$ and $\Phi(\mathcal{D}) \not\models \psi$.

Then, by the Pullback Theorem, we obtain ψ^* , a Σ_α^c sentence in the language of K such that $\mathcal{C} \models \psi^*$ and $\mathcal{D} \not\models \psi^*$, a contradiction. $\square$

The Pullback Theorem is the primary method for showing that some class K does not embed into some class K' , with only a few results obtained by other methods. Knight noted that $(LO, \cong) \not\leq_{tc} (Mod(PA), \cong)$ since there are non-isomorphic computable linear orders, while the only computable model PA can have must be the standard model. In [1] there are some non-embedding results based on the idea of Morley degree, plus calculations of index sets. Here is another result in the same vein as Corollary 1.1.4.

Corollary 1.1.5 (Fokina, Knight, Melnikov, Quinn, Safranski, [14]). *Let K and K' be classes of structures such that K contains a pair of non-isomorphic structures $\mathcal{A}$ and $\mathcal{A}'$ that satisfy the same Σ_α^c sentences for all computable α , while K' does not contain a pair with this property. Suppose further that $\omega_1^{\mathcal{A}} = \omega_1^{\mathcal{A}'} = \omega_1^{CK}$. Then $(K, \cong) \not\leq_{tc} (K', \cong)$.*

The proof of this is essentially the same as the proof of Corollary 1.1.4. It is further shown in [14] (using ideas of Friedman and Stanley from [16]) that there is a pair of non-isomorphic undirected graphs $\mathcal{A}$ and $\mathcal{A}'$ which satisfy the same computable infinitary sentences and compute no new ordinals. The class of abelian p -groups does not have such a pair of structures, and thus is not universal for $\leq_{tc}$ with respect to isomorphism. This is particularly interesting, as isomorphism for abelian p -groups is not guaranteed by sentences at any single level, showing that the conditions implied by Corollary 1.1.4 (namely that for any fixed computable α , isomorphism must not be distinguished by Σ_α^c sentences) are necessary, but not sufficient, for universality.

The Turing computable embedding is also a powerful tool for formalizing the idea of coding in structures. Sometimes we wish to transform a L -structure into some L' -structure, in such a way that certain properties are preserved. The operator present

in the definition of the embedding is the process used to code. One such application is in the area of Δ_α^0 categoricity.

Definition 2. *Let $\mathcal{A}$ be a computable structure.*

1. *$\mathcal{A}$ is Δ_α^0 categorical if for all computable copies $\mathcal{B}$ of $\mathcal{A}$, there is a Δ_α^0 isomorphism from $\mathcal{A}$ onto $\mathcal{B}$.*
2. *$\mathcal{A}$ is relatively Δ_α^0 categorical if for all copies $\mathcal{B}$ of $\mathcal{A}$, there is a $\Delta_\alpha^0(\mathcal{B})$ isomorphism from $\mathcal{A}$ onto $\mathcal{B}$.*

There has been a large body of work investigating computable categoricity. See [5],[18], and [21] for examples. In [22], using operators similar to Turing computable embeddings, Johnson, Knight, Ocasio, Tussopov, and the present author generalize results obtained by D. Hirschfeldt, B. Khoussainov, R. Shore, and A.M. Slinko in [21]. In particular, we showed the following.

Theorem 1.1.6 (Johnson, Knight, Ocasio, Tussopov, V., [22]). *For each computable ordinal $\alpha > 1$, the following classes of structures have members which are Δ_α^0 categorical but not relatively Δ_α^0 categorical:*

1. *directed graphs*
2. *partial ordering*
3. *lattices*
4. *rings*
5. *integral domains*
6. *commutative semigroups*
7. *2-step nilpotent groups*

Further, we obtained a similar result for fields (although in a fundamentally different way), complementing a result of D. Hirschfeldt, K. Kramer, R. Miller, and A. Shlapentokh from [20].

Theorem 1.1.7 (Johnson, Knight, Ocasio, Tussopov, V., [22]). *For each computable limit ordinal $\alpha > \omega$, there is a field that is Δ_α^0 categorical, but not relatively Δ_α^0 categorical.*

1.2 Computable Infinitary Equivalence

Elementary equivalence is a central notion in model theory, and it can be easily generalized to equivalences of more complex sentences. Motivated by the use of computable infinitary sentences in the context of the Pullback Theorem and its uses, we make the following definition.

Definition 3. *Two structures, $\mathcal{A}$ and $\mathcal{B}$, in the same language, are computably infinitarily Σ_α equivalent provided that $\mathcal{A}$ and $\mathcal{B}$ satisfy the same Σ_α^c sentences. In this case, we write $\mathcal{A} \sim_\alpha^c \mathcal{B}$. If $\mathcal{A} \sim_\alpha^c \mathcal{B}$ for all computable ordinals α , then we say that $\mathcal{A}$ and $\mathcal{B}$ are computably infinitarily equivalent, and we write $\mathcal{A} \sim_{\omega_1^{CK}}^c \mathcal{B}$.*

It is worthwhile to note that although computable infinitary equivalence implies isomorphism for computable structures, it does not necessarily imply isomorphism for other (non-computable) structures. For example, ω_1^{CK} (the least non-computable ordinal) and the Harrison ordering (which is a computable linear ordering of type $\omega_1^{CK} \cdot (1 + \eta)$) are computably infinitarily equivalent, but non-isomorphic. Another example is the pair of graphs used in the proof of Corollary 1.1.5 (each a family of “daisy graphs”).

CHAPTER 2

UNIVERSALITY OF ABELIAN p -GROUPS

The relation between isomorphism and computable infinitary equivalence can be explored via the Turing computable embedding. Indeed, it is natural to ask for various computable α whether there are maxima for the $\leq_{tc}$ degree structure with respect to $\sim_\alpha^c$ which are not maximal in the usual degree structure with respect to isomorphism.

2.1 Universality of ApG for $\sim_{2m}^c$

An analysis of proofs of Quinn in [34] reveals that given any computable α , there is a subclass of the class of countable reduced abelian p -groups that is universal with respect to $\sim_\alpha^c$. Let ApG_β denote the class of countable reduced abelian p -groups of length β . Then, for any even computable ordinal α , the universal subclass is ApG_β where β is the greatest computable ordinal for which $(ApG_\beta / \cong) = (ApG_\beta / \sim_\alpha^c)$. See Appendix A for information on abelian p -groups, Ulm's theorem, and the construction of Ulm invariants.

Proposition 2.1.1. *For any class K , $(K, \sim_2^c) \leq_{tc} (ApG_\omega, \sim_2^c)$.*

Proof. First fix a computable enumeration $(\psi_n)_{n \in \omega}$ of the Σ_2^c sentences in the language of K . Each ψ_n is of the form

$$\bigvee_{i \in W_{e_n}} (\exists \bar{u}_i) \bigwedge_{j \in W_{e'_n}} (\forall \bar{v}_j) \theta_{i,j}(\bar{u}_i, \bar{v}_j),$$

where θ is finitary and quantifier free. We want to build a Turing computable embedding, so we describe how the operator works. Given some input $\mathcal{A} \in K$, the operator codes the set of Σ_2^c sentences true in $\mathcal{A}$ into the Ulm sequence of the output group in such a way that

$$\mathcal{A} \models \psi_n \text{ if and only if } U_{2n}(\Phi(\mathcal{A})) = \omega.$$

In particular, $\Phi(\mathcal{A})$ will be a group of the form

$$\mathbb{Z}_p^{m_0} \oplus \mathbb{Z}_{p^2}^\omega \oplus \mathbb{Z}_{p^3}^{m_1} \oplus \mathbb{Z}_{p^4}^\omega \oplus \dots,$$

where

$$m_n = U_{2n}(\Phi(\mathcal{A})) = \begin{cases} \omega & \text{if } \mathcal{A} \models \psi_n \\ 0 & \text{if } \mathcal{A} \not\models \psi_n. \end{cases}$$

At each stage n , the truth or falsity of each sentence ψ_m , for $m \leq n$, is approximated by searching the finite set of potential witnesses $\langle i, \bar{u}_i \rangle_{i \in W_{e_m}} \leq n$ and seeing if there is a pair $(i, \bar{u}_i)$ such that $\mathcal{A} \models \theta_{i,j}(\bar{u}_i, \bar{v}_j)$ for all $\langle j, \bar{v}_j \rangle_{j \in W_{e'_m}} \leq n$. If there is such a witnessing pair, then Φ *believes* that $\mathcal{A} \models \psi_m$ up to stage n and thus builds a copy of $\mathbb{Z}_{p^{2m+1}}$ as a summand in the output group. If at stage n , it is discovered that a previously believed witness for the truth of ψ_m now fails for some $\langle j, \bar{v}_j \rangle_{j \in W_{e'_m}} \leq n$, then all previously built copies of $\mathbb{Z}_{p^{2m+1}}$ must be *trashed*. This is done by adding divisibility statements to the output group, which turn all copies of $\mathbb{Z}_{p^{2m+1}}$ into copies of $\mathbb{Z}_{p^{2m+2}}$.

Finally, at the end of each stage, the operator builds a copy of $\mathbb{Z}_{p^{2m+2}}$ as a summand in the output group for each $m \leq n$, and adds statements about all sums of elements already in the group in order to ensure that the output group is of the desired form. In the limit, if ψ_m really is true in $\mathcal{A}$, then Φ will eventually find a witness that actually does work for all $(j, \bar{v}_j)$, and will thus build infinitely many summands of the form $\mathbb{Z}_{p^{2m+1}}$ that are never trashed. If ψ_m is not true in $\mathcal{A}$, no such witness

will be found, and all summands of the form $\mathbb{Z}_{p^{2m+1}}$ are eventually trashed by being turned into summands of the form $\mathbb{Z}_{p^{2m+2}}$. Then, for any $\mathcal{A}$ and $\mathcal{B}$ in K , $\mathcal{A} \sim_2^c \mathcal{B}$ if and only if $\Phi(\mathcal{A}) \cong \Phi(\mathcal{B})$ if and only if $\Phi(\mathcal{A}) \sim_2^c \Phi(\mathcal{B})$. $\square$

In order to extend this result to obtain universal classes for $\sim_\alpha^c$ for larger computable ordinals α , we need a theorem of Khisamiev. In [28], he gives a characterization of the collection of reduced abelian p -groups of length less than ω^2 that have computable copies. We give his result below.

Theorem 2.1.2 (Khisamiev, [28]). *Let $\mathcal{G}$ be a reduced abelian p -group such that $\lambda(\mathcal{G}) < \omega^2$. Then $\mathcal{G}$ has an X -computable copy if and only if for each i such that $\omega \cdot (i+1) \leq \lambda(\mathcal{G})$, the following hold:*

- (a) $R_i = \{(n, k) : U_{\omega \cdot i + n}(\mathcal{G}) \geq k\}$ is $\Sigma_{2i+2}^0(X)$,
- (b) there is a $\Delta_{2i+1}^0(X)$ function f_i such that for all n , $f_i(n, s)$ is non-decreasing in s , with limit $f_i^*(n) \geq n$, where $U_{\omega \cdot i + f_i^*(n)}(\mathcal{G}) \neq 0$.

Further, from appropriate indices for the R_i (as a relation relative to X) and f_i (as a function relative to X), we can effectively find an index for $\mathcal{G}$ (as a structure computable relative to X).

Now, we are able to extract from [34] a universal class for $\leq_{tc}$ with respect to $\sim_{2n}^c$ for all $n \in \omega$.

Proposition 2.1.3. *For any class K , $(K, \sim_{2m}^c) \leq_{tc} (ApG_{\omega \cdot m}, \sim_{2m}^c)$.*

Proof. As before, fix a computable enumeration $(\psi_n)_{n \in \omega}$ of the Σ_{2m}^c sentences in the language of K . Given some input structure $\mathcal{A} \in K$, the operator will code the truth or falsity of the relevant set of sentences into the Ulm sequence of the output group.

Namely, we wish for $\Phi(\mathcal{A}) = \mathcal{G}$, such that $\mathcal{G} \in ApG_{\omega \cdot m}$ and

$$U_{\alpha}(\mathcal{G}) = \begin{cases} 0 & \text{if } \alpha = \omega \cdot l + 2n \text{ for } l < m - 1 \\ \omega & \text{if } \alpha = \omega \cdot l + 2n + 1 \text{ for } l < m - 1 \\ \omega & \text{if } \alpha = \omega \cdot (m - 1) + 2n \text{ and } \mathcal{A} \models \psi_n \\ 0 & \text{if } \alpha = \omega \cdot (m - 1) + 2n \text{ and } \mathcal{A} \not\models \psi_n \\ \omega & \text{if } \alpha = \omega \cdot (m - 1) + 2n + 1. \end{cases}$$

In order to use Theorem 2.1.2 to show that $\mathcal{G}$ has an $\mathcal{A}$ -computable copy, we must check that the conditions of the theorem are satisfied, namely that for each $i \leq m - 1$, the following hold:

- (a) $R_i = \{(n, k) : U_{\omega \cdot i + n}(\mathcal{G}) \geq k\}$ is $\Sigma_{2i+2}^c(\mathcal{A})$,
- (b) there is a $\Delta_{2i+1}^0(\mathcal{A})$ function f_i such that for all n , $f_i(n, s)$ is non-decreasing in s , with limit $f_i^*(n) \geq n$, where $U_{\omega \cdot i + f_i^*(n)}(\mathcal{G}) \neq 0$.

For $i < m - 1$, Condition (a) is satisfied (indeed R_i is computable), since

$$U_{\omega \cdot i + n}(\mathcal{G}) = \begin{cases} 0 & \text{if } n \text{ is even} \\ \omega & \text{if } n \text{ is odd.} \end{cases}$$

For $i = m - 1$, Condition (a) is satisfied, since

$$U_{\omega \cdot i + n}(\mathcal{G}) = \begin{cases} 0 & \text{if } n = 2k \text{ and } \mathcal{A} \models \psi_k \\ \omega & \text{if } n = 2k \text{ and } \mathcal{A} \not\models \psi_k \\ \omega & \text{if } n \text{ is even,} \end{cases}$$

and checking whether or not $\mathcal{A} \models \psi_k$ is $\Sigma_{2m}^0(\mathcal{A})$.

For $i \leq m - 1$, Condition (b) is satisfied, since $f_i(n, s) = 2n$ suffices.

Thus, $\mathcal{G} = \Phi(\mathcal{A})$ has a copy computable in $\mathcal{A}$, and this is obtained uniformly in $\mathcal{A}$ by the uniformity of Khisamiev's Theorem. $\square$

2.2 Universality of ApG for $\sim_\alpha^c$

In order to further extract maxima, we need some machinery developed by Barwise in [7] and beautifully explained in [5].

Definition 4. *Given a class K of structures in a fixed language, the standard back-and-forth relations $\leq_\beta$ on pairs $(\mathcal{A}, \bar{a})$, where $\mathcal{A} \in K$, and $\bar{a}$ is a tuple in $\mathcal{A}$, are defined as follows. For simplicity, first suppose that $\bar{a}$ in $\mathcal{A}$ and $\bar{b}$ in $\mathcal{B}$ are tuples of the same length.*

- (1) $(\mathcal{A}, \bar{a}) \leq_1 (\mathcal{B}, \bar{b})$ if and only if all finitary Σ_1 formulas true of $\bar{b}$ in $\mathcal{B}$ are also true of $\bar{a}$ in $\mathcal{A}$.
- (2) For $\alpha > 1$, $(\mathcal{A}, \bar{a}) \leq_\alpha (\mathcal{B}, \bar{b})$ if and only if, for each $\bar{d}$ in $\mathcal{B}$, and each $1 \leq \beta < \alpha$, there is a $\bar{c}$ in $\mathcal{A}$ such that $(\mathcal{B}, \bar{b}, \bar{d}) \leq_\beta (\mathcal{A}, \bar{a}, \bar{c})$.

This generalizes to tuples of different lengths as follows. For $\bar{a}$ in $\mathcal{A}$ and $\bar{b}$ in $\mathcal{B}$, $(\mathcal{A}, \bar{a}) \leq_\beta (\mathcal{B}, \bar{b})$ if and only if $|\bar{a}| \leq |\bar{b}|$ and $(\mathcal{A}, \bar{a}) \leq_\beta (\mathcal{B}, \bar{b}')$ where $\bar{b}' = \bar{b} \upharpoonright |\bar{a}|$.

When $\bar{a}$ and $\bar{b}$ are the empty tuple, it is standard to write $\mathcal{A} \leq_\beta \mathcal{B}$. The following useful characterization of the standard back-and-forth relations is due to Karp [25].

Theorem 2.2.1 (Karp, [25]). *Let $\mathcal{A}$ and $\mathcal{B}$ be countable structures, having tuples $\bar{a}$ and $\bar{b}$ such that $|\bar{a}| = |\bar{b}|$. Then for any (not necessarily computable) countable ordinal $\beta \geq 1$, the following are equivalent:*

- (1) $(\mathcal{A}, \bar{a}) \leq_\beta (\mathcal{B}, \bar{b})$,
- (2) the Σ_β formulas true of $\bar{b}$ in $\mathcal{B}$ are also true of $\bar{a}$ in $\mathcal{A}$,
- (3) the Π_β formulas true of $\bar{a}$ in $\mathcal{A}$ are also true of $\bar{b}$ in $\mathcal{B}$.

Sometimes we care about the computability of the back-and-forth relations, which leads to the following definition of Ash and Knight [5].

Definition 5. *Let $\mathcal{F} = \{\mathcal{A}_0, \mathcal{A}_1, \dots\}$ be a uniformly computable family of structures indexed by natural numbers. Representing each $\mathcal{A}_i$ by i , we say that $\mathcal{F}$ is α -friendly*

if for $\beta < \alpha$ the standard back-and-forth relations $\leq_\beta$ on the set of pairs $(i, \bar{a})$ (as a subset of $\omega \times \omega^{<\omega}$, where $\bar{a}$ is in $\mathcal{A}_i$) are c.e. uniformly in β .

In the definition above on α -friendliness, we identify the ordinals with their notations. Namely, given some fixed notation s_α for α , each $\beta < \alpha$ referred to in the definition is identified with its unique notation s_β such that $s_\beta <_{\mathcal{O}} s_\alpha$. This convention will be followed in all further definitions and proofs involving α -friendliness. The purpose of this machinery is to use the following theorem of Ash and Knight from [5] as a tool in constructing an embedding.

Theorem 2.2.2 (Ash, Knight, [5]). *Let $\mathcal{A}_0$ and $\mathcal{A}_1$ be structures such that $\mathcal{A}_0 \leq_\alpha \mathcal{A}_1$ and $\{\mathcal{A}_0, \mathcal{A}_1\}$ is α -friendly. Then for any Σ_α set S , there is a uniformly computable sequence of structures $(\mathcal{B}_n)_{n \in \omega}$ such that*

$$\mathcal{B}_n \cong \begin{cases} \mathcal{A}_0 & \text{if } n \in S \\ \mathcal{A}_1 & \text{if } n \notin S. \end{cases}$$

Often it is actually convenient to consider abstract back-and-forth relations which are *stronger* than the standard back-and-forth relations, as the calculations proving α -friendliness for the abstract case may be easier. In [34], Quinn modified results of Calvert from [9] (which, in turn depended on work of Barker in [6]) in order to say which classes Turing computably embed into ApG_{ω^2} with respect to isomorphism. We present Quinn's results on α -friendliness, with minor modifications that are needed for our context. The groups $\mathcal{G}_0^\beta$ and $\mathcal{G}_\omega^\beta$ discussed below will be used in our future coding schemes.

Definition 6. *Let $i \in \{0, \omega\}$, and $\beta \in \omega_1^{CK}$. Then $\mathcal{G}_i^\beta$ is a reduced abelian p -group*

of length $\omega \cdot (\beta + 1)$ such that

$$U_\alpha(\mathcal{G}_i^\beta) = \begin{cases} i & \text{if } \alpha = \omega \cdot \beta + 1 \\ 0 & \text{if } \alpha = \omega \cdot \gamma + 1 \text{ for } \gamma < \beta \\ \omega & \text{if } \alpha = \omega \cdot \gamma + n \text{ for } \gamma \leq \beta \text{ and even } n \\ 0 & \text{if } \alpha = \omega \cdot \gamma + n \text{ for } \gamma \leq \beta \text{ and odd } n > 1 \end{cases}$$

In the definition below, $\bar{a}$ is a tuple of elements from the group $\mathcal{G}$, and $\langle \bar{a} \rangle$ denotes the subgroup of $\mathcal{G}$ generated by $\bar{a}$.

Definition 7. Let $\mathcal{G}_i^\beta$ be the groups defined above, $i, j \in \{0, \omega\}$, and $\delta \leq \beta$. Then the abstract back-and-forth relations, $\leq_\alpha$ on these groups are defined as follows.

1. $(\mathcal{G}_i^\beta, \bar{a}) \leq_{2 \cdot \delta} (\mathcal{G}_j^\beta, \bar{b})$ if and only if
 - (a) the function f taking $\bar{a} = (a_0, \dots, a_n)$ to $\bar{b} = (b_0, \dots, b_n)$ by $f(a_k) = b_k$ extends to an isomorphism $\hat{f} : \langle \bar{a} \rangle \rightarrow \langle \bar{b} \rangle$,
 - (b) for all $b \in \bar{b}$ and $a = f(b)$, either $h(a) = h(b) < \omega \cdot \delta$, or $h(a), h(b) \geq \omega \cdot \delta$,
 - (c) for all $\gamma \leq \omega \cdot \delta$, $U_\gamma(\mathcal{G}_i^\beta) = U_\gamma(\mathcal{G}_j^\beta)$.
2. $(\mathcal{G}_i^\beta, \bar{a}) \leq_{2 \cdot \delta + 1} (\mathcal{G}_j^\beta, \bar{b})$ if and only if
 - (a) the function f taking $\bar{a} = (a_0, \dots, a_n)$ to $\bar{b} = (b_0, \dots, b_n)$ by $f(a_k) = b_k$ extends to an isomorphism $\hat{f} : \langle \bar{a} \rangle \rightarrow \langle \bar{b} \rangle$,
 - (b) for all $b \in \bar{b}$ and $a = f(b)$, either $h(a) = h(b) < \omega \cdot \delta$ or $h(b) \geq \omega \cdot \delta$ and $h(a) \geq \min\{h(b), \omega \cdot \delta + \omega\}$,
 - (c) for all $\gamma \leq \omega \cdot \delta$, $U_\gamma(\mathcal{G}_i^\beta) = U_\gamma(\mathcal{G}_j^\beta)$.

In order to use Theorem 2.2.2, we follow the proofs in [34] showing that these abstract back-and-forth relations actually do have the back-and-forth property, that $\{\mathcal{G}_0^\beta, \mathcal{G}_\omega^\beta\}$ is $2(\beta + 1)$ -friendly, and that $\mathcal{G}_\omega^\beta \leq_{2(\beta+1)} \mathcal{G}_0^\beta$.

Lemma 2.2.3 (Quinn, [34]). *The abstract back-and-forth relations in Definition 7 satisfy the back-and-forth property. Namely, let $(\mathcal{G}_i^\beta, \bar{a}) \leq_\alpha (\mathcal{G}_j^\beta, \bar{b})$ (in the sense of Definition 7). Then for any $\lambda < \alpha$ and any $\bar{d} \in \mathcal{G}_j^\beta$, there is a $\bar{c} \in \mathcal{G}_i^\beta$ such that $(\mathcal{G}_j^\beta, \bar{b}, \bar{d}) \leq_\lambda (\mathcal{G}_i^\beta, \bar{a}, \bar{c})$.*

Proof. Based on the definition of the abstract back-and-forth relations, there are two cases.

Case 1: $(\mathcal{G}_i^\beta, \bar{a}) \leq_{2 \cdot (\delta+1)} (\mathcal{G}_j^\beta, \bar{b})$

First, note that we will only be using Theorem 2.2.2 in the case where $\delta \leq \beta$ (recall that $\mathcal{G}_i^\beta$ and $\mathcal{G}_j^\beta$ have length $\omega \cdot (\beta + 1)$, and that in Definition 7, $\delta \leq \beta$). Fix some $\lambda \leq 2 \cdot \delta + 1 < 2 \cdot (\delta + 1)$, and $\bar{d} \in \mathcal{G}_j^\beta$. By the inductive definition of the back-and-forth property, we may assume that $\lambda = 2 \cdot \delta + 1$. Then we must find some $\bar{c} \in \mathcal{G}_i^\beta$ such that $(\mathcal{G}_j^\beta, \bar{b}, \bar{d}) \leq_\lambda (\mathcal{G}_i^\beta, \bar{a}, \bar{c})$. Note that since $\lambda = 2 \cdot \delta + 1$, this falls under Case 2 of Definition 7, so we check that the conditions are satisfied.

- (a),(b) Pick some $d \in \bar{d} \setminus \langle \bar{b} \rangle$. In order to satisfy Conditions (a) and (b), we must find a $c \in \mathcal{G}_i^\beta$ such that $h(d) = h(c) < \omega \cdot (\delta + 1)$ or that $h(c) \geq \min\{h(d), \omega \cdot (\delta + 1) + \omega\}$. Finding this c breaks into two sub-cases. Consider the sub-case where one of $\delta = \beta$, $i = 0$, $j = \omega$, or $h(d) = \omega \cdot \beta + 1$ is false. We can find a c having the desired properties, as $\mathcal{G}_i^\beta$ and $\mathcal{G}_j^\beta$ have the same Ulm invariants except at $\omega \cdot \beta + 1$. In the sub-case where $\delta = \beta$, $i = 0$, $j = \omega$, or $h(d) = \omega \cdot \beta + 1$, we take $c \in \mathcal{G}_0^\beta$ to be an element witnessing the fact that $U_{\omega \cdot \beta + 2}(\mathcal{G}_0^\beta) = \omega$, which must exist. Finally, an analysis of Kaplansky's proof of Ulm's theorem in [24] verifies that $f : (\bar{a}, \bar{c}) \rightarrow (\bar{b}, \bar{d})$ can be extended to $\hat{f} : \langle \bar{a}, \bar{c} \rangle \rightarrow \langle \bar{b}, \bar{d} \rangle$. Thus Conditions (a) and (b) are satisfied.
- (c) The only Ulm invariant on which $\mathcal{G}^{m_i}$ and $\mathcal{G}^{m_j}$ differ is $U_{\omega \cdot m + 1}$, and $\delta \leq m - 1$, so Condition (c) is satisfied.

Case 2: $(\mathcal{G}^{m_i}, \bar{a}) \leq_{2 \cdot (\delta+1)+1} (\mathcal{G}^{m_j}, \bar{b})$

As before, we have $\delta + 1 \leq m$, and it is sufficient to verify that there are witnesses $\bar{c}$ for the back-and-forth property for $\lambda = 2 \cdot \delta + 1$. So, we must find some $\bar{c} \in \mathcal{G}^{m_i}$ such that $(\mathcal{G}_j^\beta, \bar{b}, \bar{d}) \leq_\lambda (\mathcal{G}_i^\beta, \bar{a}, \bar{c})$. This falls under case 1 of Definition 7, so we check that the conditions are satisfied.

- (a),(b) These conditions are proven similarly to those in Case 1. Suppose that $\delta = \beta$, $i = 0$, $j = \omega$. If we are given some $\bar{d}$ and need to find a $\bar{c}$ witnessing that $(\mathcal{G}_\omega^\beta, \bar{b}, \bar{d}) \leq_{2 \cdot (\delta+1)} (\mathcal{G}_0^\beta, \bar{a}, \bar{c})$, then we use the fact that all $U_{\omega \cdot \beta + 2n}(\mathcal{G}_0^\beta)$ are infinite. In particular, we need a $c \in \mathcal{G}_0^\beta$ such that $h(c) \geq \omega \cdot \beta$, the existence of which is guaranteed by the Ulm sequence. Again, Kaplansky's proof of Ulm's theorem shows that we may extend a height-preserving mapping to an isomorphism of the subgroup.

(c) Again, the groups agree on all Ulm invariants other than the $\omega \cdot m + 1$ st.

□

Lemma 2.2.4 (Quinn, [34]). *Let $\mathcal{G}_0^\beta$ and $\mathcal{G}_\omega^\beta$ be as above. Then $\{\mathcal{G}_0^\beta, \mathcal{G}_\omega^\beta\}$ is a $2(\beta+1)$ -friendly family.*

Proof. We must show $\mathcal{G}_0^\beta$ and $\mathcal{G}_\omega^\beta$ have computable copies on which the back-and-forth relations $\leq_\eta$, $\eta < 2(\beta+1)$ are computably enumerable uniformly in η . Let $(A_n)_{n \in \omega}$ be a computable partition of ω . We build computable copies of $\mathcal{G}_i^\beta$, $i \in \{0, \omega\}$ such that for every $a \in |\mathcal{G}_i^\beta|$, $a \in A_{\langle \gamma, t \rangle}$ if and only if $h(a) = \omega \cdot \gamma + t$ (recall that we are identifying γ with its notation $s_\gamma <_{\mathcal{O}} s_\beta$ where s_β is the fixed notation for β). To do this, first note that $2(\beta+1)$ is computable, so we can fix a computable enumeration of all computable ordinals less than it. Based on this enumeration, we build $\mathcal{G}_i^\beta$, at each step deciding we want an element of some height $\omega \cdot \gamma + t$, and picking an $a \in A_{\gamma, t}$ to witness this desired height. This is sufficient, as checking whether $(\mathcal{G}_i^\beta \bar{a}) \leq_{2(\beta+1)} (\mathcal{G}_j^\beta \bar{b})$ only requires checking the conditions (a), (b) and (c), which only requires knowing the heights (now computable in our groups), and whether a height-preserving function on tuples can be extended to a partial isomorphism. This is a c.e. process. □

Lemma 2.2.5 (Quinn, [34]). *Let $\mathcal{G}_0^\beta$ and $\mathcal{G}_\omega^\beta$ be as above. Then $\mathcal{G}_\omega^\beta \leq_{2(\beta+1)} \mathcal{G}_0^\beta$.*

Proof. Given some $\bar{a}$ in $\mathcal{G}_0^\beta$, we must find a $\bar{b}$ in $\mathcal{G}_\omega^\beta$ such that $(\mathcal{G}_0^\beta, \bar{a}) \leq_{2\beta+1} (\mathcal{G}_\omega^\beta, \bar{b})$. Since $U_{\omega, \beta+1}(\mathcal{G}_0^\beta) = 0$, there is no difficult case here. Given any $\bar{a}$, we can find some tuple $\bar{b}$ of $\mathcal{G}_\omega^\beta$ satisfying all of the same $\Pi_{2\beta+1}^c$ formulas in $\mathcal{G}_\omega^\beta$ that $\bar{a}$ does in $\mathcal{G}_0^\beta$. In particular, given $a_0 \in \bar{a}$, we choose $b_0 \in \mathcal{G}_\omega^\beta$ such that $h(a) = h(b)$. We then choose set each b_i to be any element of $\mathcal{G}_\omega^\beta \setminus \langle b_0, \dots, b_{i-1} \rangle$ such that $h(b_i) = h(a_i)$. □

One final ingredient will be needed in the proof, namely that the direct sum of infinitely many groups can be simulated by a Turing computable operator. This is not surprising, as given a collection of groups, we *know* how to construct the sum.

Lemma 2.2.6. *There is a computable operator $\Phi = \varphi_e$ such that, given an a computable ordering $(I, <_I)$, and an I -indexed family of groups $(\mathcal{G}_i)_{i \in I} = \{\langle i, \psi \rangle : \psi \in D(\mathcal{G}_i)\}$,*

$$\Phi((\mathcal{G}_i)_{i \in I}) = \varphi_e^{(\mathcal{G}_i)_{i \in \omega}} = \chi_{\mathcal{G}},$$

where $\mathcal{G} \cong \oplus_{i \in \omega} \mathcal{G}_i$.

Proof. In this proof, we use notation for abelian groups (0_i and $+_i$) but really the lemma is true for non-abelian groups as well. Let (i, a) denote an element a of G_i . The universe of our output structure consists of finite sequences of elements from the groups. Namely, each element of the universe is thought of as a sequence $((i_j, a_j))_{j \in I}$, where a_j is in G_{i_j} , and a_j is not the identity 0_{i_j} in $\mathcal{G}_{i_j}$. Since these are really just finite strings, there is no difficulty coding them as natural numbers in the output structure. We let $\langle (i_0, a_{i_0}), (i_1, a_{i_1}), \dots, (i_n, a_{i_n}) \rangle$ where $i_0 <_I i_1 <_I \dots <_I i_n$ represent the sequence having precisely $a_{i_0}, \dots, a_{i_n}$ non-identity. Thus the identity of the output group will be represented by the code for the empty tuple, $\langle \rangle = 1$.

Now, we build the operator Φ . Given an input structure,

$$(G_i)_{i \in I} = \{\langle i, \varphi \rangle : i \in I, \varphi \in D(G_i)\},$$

the operator begins by searching for elements of the various groups by running along the oracle, looking for statements of the form ' $\langle i, a = a \rangle$ ' for various i . When we see one, we put the (code for the) corresponding (i, a_i) into the universe of the output. At some point in the construction of the output, say we have put two elements into the universe. Then we must put their product (group operation) into the universe as well. Namely, we need to define the group operation on the output structure. Suppose the elements are of the forms $\langle (i_0, a_{i_0}), (i_1, a_{i_1}), \dots, (i_n, a_{i_n}) \rangle$ and $\langle (j_0, b_{j_0}), (j_1, b_{j_1}), \dots, (j_m, b_{j_m}) \rangle$. Then we know what the sum of these two elements *should* be. We first search and find all pairs of matching indices from the tuples; for

all pairs $i_c = j_d$. Then we compute $a_{i_c} +_{i_c} b_{j_d}$ where $+_i$ is the group operation from G_i . The important fact is that we need only compute *finitely* many operations, so we may search the oracle for the answer to each question of the form ‘what is $a_i +_i b_i$?’ When we find each of the finitely many answers, we code the tuple, consisting of these answers, and the collection of elements from both tuples which were not operands into the universe of the output structure.

Two difficulties may arise here. First, if $i_c = j_d$, and $a_{i_c} +_{i_c} b_{j_d} = 0_{i_c}$, then we do not put $(i_c, 0_{i_c})$ into the output tuple (since we only want non-identity elements in the tuple). Secondly, when asking the oracle ‘what is $a_i +_i b_i$?’ we may receive as an answer some element r_i that is not yet coded in the universe of the output structure. If this happens, we code it into the output as $\langle(i, r_i)\rangle$, and proceed. In the limit, we produce a group isomorphic to $\mathcal{G}$. $\square$

This lemma seems related to work of Feferman and Vaught in [13], where they studied how abstract sums of structures preserve decidability. It would be interesting to see which of their theorems transfer to the context of Turing computable operators. Armed with the theorems above, we are now able to extract a universal class from a proof of Quinn in [34].

Proposition 2.2.7. *For any class K , $(K, \sim_\omega^c) \leq_{tc} (ApG_{\omega^2}, \sim_\omega^c)$.*

Proof. First note that $\mathcal{A} \sim_\omega^c \mathcal{A}'$ precisely when $\mathcal{A} \sim_n^c \mathcal{A}'$ for all n . Thus, we need only build an operator that codes the collection of all arithmetical sentences true in the input structure $\mathcal{A}$ into the output group $\mathcal{G}$. Fix a computable enumeration $(\psi_{\langle m, n \rangle})_{\langle m, n \rangle \in \omega}$ where ψ_i is the n th sentence in a (fixed) computable enumeration of all Σ_m^c sentences in the language of the input class K precisely when $i = \langle m, n \rangle$. Note that this enumeration can be found in such a way that each ψ_i is at most Σ_{2i+2}^c . For each ψ_i the operator builds a group $\mathcal{G}_{(i)}$ of length $\omega \cdot (i + 1)$ which codes the truth or

falsity of ψ_i as follows.

$$U_\alpha(\mathcal{G}_{(i)}) = \begin{cases} \omega & \text{if } \alpha = \omega \cdot i + 1 \text{ and } \mathcal{A} \models \psi_i \\ 0 & \text{if } \alpha = \omega \cdot i + 1 \text{ and } \mathcal{A} \not\models \psi_i \\ \omega & \text{if } \alpha = \omega \cdot j + 2k + 1 \text{ where } j \leq i, 0 < k \\ 0 & \text{if } \alpha = \omega \cdot j + 2k \text{ where } j \leq i, 0 < k. \end{cases}$$

Then, the final output of the operator will be $\mathcal{G} := \oplus \mathcal{G}_{(i)}$, which is computable by Lemma 2.2.6, will have an Ulm sequence coding the truth or falsity of each ψ_i in the the $(\omega \cdot i + 1)$ st member of the sequence, 0 at each other odd member of the sequence, and ω at every even member of the sequence. In order to actually build the operator, we use Theorem 2.2.2 and Lemma 2.2.6.

Note that checking whether or not $\mathcal{A} \models \psi_{\langle m, n \rangle}$ is $\Sigma_m(\mathcal{A})$. As the above lemmas on back-and-forth relations show that the coding structures $\mathcal{G}_{(i)} = \mathcal{G}_0^m$ or $\mathcal{G}_\omega^m$ satisfy the conditions of Theorem 2.2.2 uniformly. So the lemma guarantees that we build the correct coding group, based on whether $\mathcal{A} \models \psi_{\langle m, n \rangle}$. Then we use Lemma 2.2.6 to create our output:

$$\Phi(\mathcal{A}) = \mathcal{G} = \oplus \mathcal{G}_{(i)}.$$

This group has length ω^2 and the desired Ulm sequence

$$U_\alpha(\mathcal{G}) = \begin{cases} \omega & \text{if } \alpha = \omega \cdot i + 1 \text{ and } \mathcal{A} \models \psi_i \\ 0 & \text{if } \alpha = \omega \cdot i + 1 \text{ and } \mathcal{A} \not\models \psi_i \\ \omega & \text{if } \alpha = \omega \cdot j + 2k \text{ where } j \leq i, 0 < k \\ 0 & \text{if } \alpha = \omega \cdot j + 2k + 1 \text{ where } j \leq i, 0 < k. \end{cases}$$

□

With minor modification, this proof extends to groups of higher computable ordinal length, and yields classes that are universal for $\sim_\alpha^c$ for any *particular* computable

α .

Proposition 2.2.8 (Quinn, [34]). *For any class K ,*

$$(K, \sim_\beta^c) \leq_{tc} (ApG_\alpha, \sim_\beta^c),$$

for $\alpha = \omega \cdot \beta$.

Proof. First, fix the computable sequence $(\beta_n)_{n \in \omega}$ of computable ordinals with limit β , given by the notation for β . Note that $\mathcal{A} \sim_\beta^c \mathcal{A}'$ precisely when $\mathcal{A} \sim_{\beta_n}^c \mathcal{A}'$ for all n . Thus, we need only build an operator that codes the collection of all $\Sigma_{\beta_n}^c$ sentences true in the input structure $\mathcal{A}$ into the output group $\mathcal{G}$ for all n . Fix a computable enumeration $(\psi_{\langle m, n \rangle})_{\langle m, n \rangle \in \omega}$ where ψ_i is the n th sentence in a (fixed) computable enumeration of all $\Sigma_{\beta_m}^c$ sentences, in the language of the input class K precisely when $i = \langle m, n \rangle$. Note that this enumeration can be found in such a way that each ψ_i is at most $\Sigma_{2 \cdot \beta_m + 2}^c$. For each ψ_i , the operator builds a group $\mathcal{G}_{(i)}$, of length $\omega \cdot (\beta_m + 1)$, that codes the truth or falsity of ψ_i as follows.

$$U_\alpha(\mathcal{G}_{(i)}) = \begin{cases} \omega & \text{if } \alpha = \omega \cdot \beta_m + 1 \text{ and } \mathcal{A} \models \psi_i \\ 0 & \text{if } \alpha = \omega \cdot \beta_m + 1 \text{ and } \mathcal{A} \not\models \psi_i \\ \omega & \text{if } \alpha = \omega \cdot \beta_j + 2k \text{ where } \beta_j \leq \beta_m, 0 < k \\ 0 & \text{if } \alpha = \omega \cdot \beta_j + 2k + 1 \text{ where } \beta_j \leq \beta_m, 0 < k. \end{cases}$$

Then, the final output of the operator will be $\mathcal{G} := \oplus \mathcal{G}_{(i)}$, which is computable by Lemma 2.2.6, will have an Ulm sequence coding the truth or falsity of each ψ_i in the $(\omega \cdot \beta_m + 1)$ st member of the sequence, and having 0 at each other odd member of the sequence, and ω at every even member of the sequence. In order to actually build the operator, we use Theorem 2.2.2 and Lemma 2.2.6.

Note that checking whether or not $\mathcal{A} \models \psi_{\langle m, n \rangle}$ is $\Sigma_{\beta_m}(\mathcal{A})$. As the above lemmas on back-and-forth relations show, for each i , the coding structures $\mathcal{G}_{(i)} = \mathcal{G}_0^{\beta_m}$ or $\mathcal{G}_\omega^{\beta_m}$

satisfy the conditions of Theorem 2.2.2. So the lemma guarantees that we build the correct coding group, based on whether $\mathcal{A} \models \psi_{\langle m, n \rangle}$. Then, we use Lemma 2.2.6 to create our output:

$$\Phi(\mathcal{A}) = \mathcal{G} = \oplus \mathcal{G}_{(i)}.$$

This group has length $\omega \cdot \beta$ and the desired Ulm sequence

$$U_\alpha(\mathcal{G}) = \begin{cases} \omega & \text{if } \alpha = \omega \cdot \beta_m + 1 \text{ and } \mathcal{A} \models \psi_i \\ 0 & \text{if } \alpha = \omega \cdot \beta_m + 1 \text{ and } \mathcal{A} \not\models \psi_i \\ \omega & \text{if } \alpha = \omega \cdot \beta_j + 2k \text{ where } \beta_j \leq \beta_m, 0 < k \\ 0 & \text{if } \alpha = \omega \cdot \beta_j + 2k + 1 \text{ where } \beta_j \leq \beta_m, 0 < k. \end{cases}$$

□

Note that until Proposition 2.2.8, all embeddings were “tight” in the sense that the output groups had isomorphism equivalent to $\sim_\alpha^c$ for the particular α for which they were universal.

CHAPTER 3

UNIFORMITY AND THE RECURSION THEOREM

In the main propositions in the previous chapter, we modified results of Quinn from [34] to show that certain subclasses of the class of abelian p -groups are universal for our relations $\sim_\alpha^c$ under Turing computable embedding. These subclasses, ApG_λ , are natural in the sense that isomorphism often coincides with computable infinitary Σ_α equivalence on the classes (not in Proposition 2.2.8). Here we will aim for sharper results. In particular, given some computable α , we can find a class of abelian p -groups which is universal for $\sim_\alpha^c$ by looking at the class of groups whose Ulm sequences are completely described by Σ_α^c sentences. In each case, the operator witnessing the universality codes the set of Σ_α^c sentences true in the input structure into the Ulm sequence of the output structure. In this chapter, we further refine these operators, exploit their uniformity, and use transfinite induction via the recursion theorem to show that there is a single program that computes the length of the groups we need together with an index for an operator witnessing an embedding.

3.1 Transfinite Induction and the Recursion Theorem

Transfinite induction on ordinals is a proof technique used to show that all ordinals have some property, P . It is carried out by proving the following three facts.

1. The ordinal 0 has property P .
2. If α has property P , then $\alpha + 1$ also has property P .
3. If λ is a limit ordinal, and all $\beta < \lambda$ have property P , then λ also has property P .

Transfinite induction on ordinals is easily translated into the context of $\mathcal{O}$, Kleene's system of ordinal notations. If we wish to show that all elements of $\mathcal{O}$ satisfy some property P , we prove the following three facts.

1. 1 has property P .
2. For $b \in \mathcal{O}$, if b has property P , then so does 2^b .
3. For $a = 3 \cdot 5^e \in \mathcal{O}$, if for all n , $\varphi_e(n)$ has property P , then a also has property P .

This proof technique is called *recursive transfinite induction on ordinal notation*, and it justifies a scheme, due to Kleene, for defining a partial computable function with domain a superset of $\mathcal{O}$. In order to ensure that a partial computable function f , with domain a superset of $\mathcal{O}$ is well-defined, it is sufficient to specify the following.

1. Define $f(1)$.
2. Say how $f(2^b)$ is computed from $f(b)$.
3. Say how $f(3 \cdot 5^e)$ is computed from the sequence $(f(\varphi_e(n)))_{n \in \omega}$.

Kleene's original collection of results proved using his Recursion Theorem includes the following theorem showing that there is a partial computable function that simulates ordinal addition.

Theorem 3.1.1 (Kleene, [29]). *There is a partial computable function $+_{\mathcal{O}}$ such that for all $a, b \in \mathcal{O}$,*

- (a) $+_{\mathcal{O}}(a, b) \in \mathcal{O}$,
- (b) $|a|_{\mathcal{O}} + |b|_{\mathcal{O}} = | +_{\mathcal{O}}(a, b) |_{\mathcal{O}}$,
- (c) if $b \neq 1$, then $a <_{\mathcal{O}} +_{\mathcal{O}}(a, b)$.

We will write $a +_{\mathcal{O}} b$ instead of $+_{\mathcal{O}}(a, b)$. This theorem is proven by giving a recursive definition of ordinal addition, and translating it to the context of $\mathcal{O}$, specifying that

1. $a +_{\mathcal{O}} 1 = a$,

2. $a +_{\mathcal{O}} 2^b = 2^{a+_{\mathcal{O}} b}$,
3. $a +_{\mathcal{O}} 3 \cdot 5^e = 3 \cdot 5^{e'}$, where $\varphi_{e'}(n) = a +_{\mathcal{O}} \varphi_e(n)$.

Kleene's Recursion Theorem then tells us that this is sufficient. By defining $+_{\mathcal{O}}$ for input 1, we have obtained a collection of indices of partial computable functions that “computes $+_{\mathcal{O}}$ up to notation 1.” Then, given any index for a partial computable function that “computes $+_{\mathcal{O}}$ up to notation b ,” we are able to obtain an index for a new partial computable function, extending the first, that computes $+_{\mathcal{O}}$ up to notation 2^b . Finally, given a computable sequence, f_n , of indices for partial computable functions, each of which computes $+_{\mathcal{O}}$ up to notation $\varphi_e(n)$ (where $\varphi_e(0) <_{\mathcal{O}} \varphi_e(1) <_{\mathcal{O}} \dots$), we can obtain an index for a partial computable function that acts as a limit function, extending those in the sequence. This process of obtaining indices is computable, so by the Recursion Theorem, it has a fixed point. This fixed point is an index for $+_{\mathcal{O}}$.

The argument described above lies at the heart of every proof that a definition by recursive transfinite induction on ordinal notation gives a partial computable function with a desired set of properties. The following theorem from [5] summarizes this method, and justifies definition by transfinite induction on ordinal notation.

Theorem 3.1.2. *(a) Let g be a partial computable function defined (at least) on the pairs (b, u) where $b \in \mathcal{O}$ and φ_u is a function with domain $<_{\mathcal{O}}\text{-pred}(b)$ (the collection of elements of $\mathcal{O}$ which are $<_{\mathcal{O}}$ b). Then there is a partial computable function f such that for $b \in \mathcal{O}$, $f(b) = g(b, u)$, where φ_u is the restriction of f to $<_{\mathcal{O}}\text{-pred}(b)$.*

(b) Let g be a partial computable function defined on the triples (a, b, u) where $a, b \in \mathcal{O}$ and φ_u is a function with domain $\{(a, d) : d <_{\mathcal{O}} b\}$. Then there is a partial computable function f such that for all a and b in $\mathcal{O}$,

$$f(a, b) = g(a, b, u),$$

where φ_u is the restriction of f to $\{(a, d) : d <_{\mathcal{O}} b\}$.

In order to obtain a result on embeddings and countable reduced abelian p -groups, we need a few lemmas. First, we imitate the proof of Theorem 3.1.1 in order to prove

that ordinal multiplication is also simulated on $\mathcal{O}$ by a partial computable function. This result is stated in [36] as an exercise.

Theorem 3.1.3 (H. Rogers, [36]). *There is a partial computable function $\cdot_{\mathcal{O}}$ such that for all $a, b \in \mathcal{O}$,*

- (a) $\cdot_{\mathcal{O}}(a, b) \in \mathcal{O}$,
- (b) $|a|_{\mathcal{O}} \cdot |b|_{\mathcal{O}} = |\cdot_{\mathcal{O}}(a, b)|_{\mathcal{O}}$,
- (c) if $b \neq 1$ or 2 , then $a <_{\mathcal{O}} \cdot_{\mathcal{O}}(a, b)$.

Again, the proof is by transfinite induction on ordinal notation. We give a recursive definition of ordinal multiplication and translate it to the context of $\mathcal{O}$ to state that

- 1. $\cdot_{\mathcal{O}}(a, 0) = 0$,
- 2. $\cdot_{\mathcal{O}}(a, 2^b) = \cdot_{\mathcal{O}}(a, b) +_{\mathcal{O}} a$,
- 3. $\cdot_{\mathcal{O}}(a, 3 \cdot 5^e) = \varphi_{e'} = 3 \cdot 5^{e'}$, where $\varphi_{e'}(n) = \cdot_{\mathcal{O}}(a, \varphi_e(n))$.

Then, by the Recursion Theorem, there is a partial computable function defined on $\mathcal{O}$ satisfying Properties 1, 2, and 3.

3.2 Operator Agreement

Now, we wish to show that there is a partial computable function that takes as input the notation for an ordinal α , and outputs a notation for β , so that the class of abelian p -groups of length β lies on top for $\sim_{\alpha}^c$ under Turing computable embedding. Analysis of Propositions 2.1.1, 2.1.3, 2.2.7, and 2.2.8 reveals a pattern we will exploit in defining our desired partial computable function. In order to exploit this pattern, we must refine the operators witnessing universality so that they do not interfere with one another. We formalize this notion of ‘non-interfering operators’ with the following definition.

Definition 8.

1. Suppose that Φ and Ψ are Turing computable operators taking as inputs structures in a class K , and outputting countable reduced abelian p -groups. Then Φ and Ψ agree up to α if for all $\mathcal{A} \in K$, and all $\beta < \alpha$, $U_\beta(\Phi(\mathcal{A})) = U_\beta(\Psi(\mathcal{A}))$.
2. Suppose there are γ and η such that for all $\mathcal{A} \in K$, $\lambda(\Phi(\mathcal{A})) = \gamma$ and $\lambda(\Psi(\mathcal{A})) = \eta$. Let $\alpha = \min\{\gamma, \eta\}$. Then Φ and Ψ agree if they agree up to α .
3. A collection $(\Phi_i)_{i \in I}$ of Turing computable operators is said to be agreeable if every pair of operator from the class agree.

Note that when operators Φ and Ψ agree, they do not necessarily output the same group, as one may output a group of larger length. For future results, we need to construct a class of agreeable operators $(\Phi_\alpha)_{\alpha < \omega_1^c K}$, where each Φ_α witnesses that the countable reduced abelian p -groups are universal for $\sim_\alpha^c$.

Lemma 3.2.1. *There is an agreeable family of embeddings $(\Phi_{2m})_{m \in \omega}$ such that each Φ_{2m} witnesses that for any class K , $(K, \sim_{2m}^c) \leq_{tc} (ApG_{\omega \cdot m}, \sim_{2m}^c)$.*

Proof. We modify the embeddings constructed in Proposition 2.1.3, to construct the new embeddings which agree. Let Ψ_{2n} denote the embedding from Proposition 2.1.3 witnessing that $ApG_{\omega \cdot n}$ is universal for $\sim_{2n}^c$. Then, given an $\mathcal{A} \in K$, we define

$$\Phi_{2m}(\mathcal{A}) = \oplus_{n \leq m} \Psi_{2n}(\mathcal{A}).$$

Let $(\psi_{\langle i, j \rangle})_{\langle i, j \rangle \in \omega}$ be a computable enumeration of all Σ_{2m}^c sentences in the language of K , such that each ψ_i is Σ_{2j}^c for $0 < j \leq m$. This output group has length $\omega \cdot m$ and codes the collection of all Σ_{2j}^c sentences true in $\mathcal{A}$ for all $j \leq m$ by

$$U_\alpha(\Phi_{2m}(\mathcal{A})) = \begin{cases} \omega & \text{if } \alpha = \omega \cdot (j - 1) + 2i \text{ and } \mathcal{A} \models \psi_{\langle i, j \rangle} \\ 0 & \text{if } \alpha = \omega \cdot (j - 1) + 2i \text{ and } \mathcal{A} \not\models \psi_{\langle i, j \rangle} \\ \omega & \text{if } \alpha = \omega \cdot (j - 1) + 2i + 1 \\ 0 & \text{otherwise.} \end{cases}$$

Then, note that

$$\Phi_{2m}(\mathcal{A}) = \oplus_{n \leq m} \Phi_{2n}(\mathcal{A}),$$

which ensures that any two Φ_{2m} and Φ_{2n} agree. $\square$

We now consider the operator which is the limit of the Φ_{2m} from the previous Lemma.

Lemma 3.2.2. *There is an agreeable family of embeddings Φ_α , $\alpha \in \{2, 4, 6, \dots, \omega\}$ such that*

1. Φ_ω witnesses that for any class K , $(K, \sim_\omega^c) \leq_{tc} (ApG_{\omega^2}, \sim_\omega^c)$,
2. each Φ_{2m} witnesses that for any class K , $(K, \sim_{2m}^c) \leq_{tc} (ApG_{\omega \cdot m}, \sim_{2m}^c)$.

Proof. Let Φ_{2m} be as in Lemma 3.2.1, and define

$$\Phi_\omega(\mathcal{A}) = \oplus_{m \in \omega} \Phi_{2m}(\mathcal{A}).$$

Then $\Phi_\omega(\mathcal{A})$ codes the truth or falsity of all Σ_{2m}^c sentences in $\mathcal{A}$ for all m . Thus, $\mathcal{A} \sim_\omega^c \mathcal{B}$ if and only if $\Phi_\omega(\mathcal{A}) \sim_\omega^c \Phi_\omega(\mathcal{B})$, and all Φ_α agree for $\alpha \in \{2, 4, 6, \dots, \omega\}$. $\square$

These techniques extend to find refinements for the operators witnessing universality of groups for $\sim_\alpha^c$ for arbitrarily large computable α as well. As in the proofs of Lemmas 3.2.1 and 3.2.2, at successor levels we manually code in the truth or falsity of the new collection of sentences, while retaining all previous Ulm invariants, while at limit levels we take the sum of the groups produced by previously constructed operators. In order to do the manual coding at the successor ordinals though, we must use Theorem 2.2.2, and a strategy similar to that used in the proof of Theorem 2.2.8. We first prove that we can take the limit of an agreeable ω -sequence of operators as in Lemma 3.2.2.

Proposition 3.2.3. *Suppose that $(\Phi_\alpha)_{\alpha \leq \omega \cdot \beta + 2n, n \in \omega}$ (for some fixed computable β) is an agreeable family of Turing computable operators such that each Φ_α witnesses that $(K, \sim_\alpha^c) \leq_{tc} (ApG_{g(\alpha)}, \sim_\alpha^c)$, where*

$$g(\omega \cdot \rho + 2m) = \omega^2 \cdot \rho + \omega \cdot m.$$

Fix an enumeration $(\psi_{\langle \gamma, j \rangle})_{\gamma < \omega \cdot (\beta+1)}$ of the computable infinitary sentences in the language of K such that each $\psi_{\langle \gamma, j \rangle}$ is the j th Σ_γ^c sentence in a fixed enumeration. Suppose further that each $\Phi_{\omega \cdot \rho + 2p}$ codes by

$$U_\eta(\Phi_{\omega \cdot \rho + 2p}(\mathcal{A})) = \begin{cases} \omega & \text{if } \eta = \omega^2 \cdot \rho + \omega \cdot m + 2j \text{ for } m < p \text{ and } \mathcal{A} \models \psi_{\langle \omega \cdot \beta + 2m, j \rangle} \\ 0 & \text{if } \eta = \omega^2 \cdot \rho + \omega \cdot m + 2j \text{ for } m < p \text{ and } \mathcal{A} \not\models \psi_{\langle \omega \cdot \beta + 2m, j \rangle} \\ \omega & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j \text{ for } \delta < \rho \text{ and } \mathcal{A} \models \psi_{\langle \omega \cdot \delta + 2m, j \rangle} \\ 0 & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j \text{ for } \delta < \rho \text{ and } \mathcal{A} \not\models \psi_{\langle \omega \cdot \delta + 2m, j \rangle} \\ \omega & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j + 1 < \omega^2 \cdot \rho + \omega \cdot p \\ 0 & \text{otherwise.} \end{cases}$$

Then there is a Turing computable operator, $\Phi_{\omega \cdot (\beta+1)}$ such that $(\Phi_\alpha)_{\alpha \leq \omega \cdot (\beta+1)}$ is an agreeable family, and $\Phi_{\omega \cdot (\beta+1)}$ witnesses that

$$(K, \sim_{\omega \cdot (\beta+1)}^c) \leq_{tc} (ApG_{g(\omega^2 \cdot \beta + \omega)}, \sim_{\omega \cdot (\beta+1)}^c).$$

Proof. We produce an operator $\Phi_{\omega \cdot (\beta+1)}$ that agrees with all Φ_α for $\alpha < \omega \cdot (\beta+1)$. Define $\Phi_{\omega \cdot (\beta+1)}$ by

$$\Phi_{\omega \cdot (\beta+1)}(\mathcal{A}) = \oplus_{\alpha < \omega \cdot (\beta+1)} (\mathcal{A}).$$

This has the desired properties. □

For the case when we extend an agreeable family that works up to α to an agreeable family that works up to $\alpha + 2$, we will need to use Theorem 2.2.2 with a

particular pair of structures, designed to be agreeable with embeddings at lower levels. This involves computing the back and forth relations once again. Luckily, despite the fact that we are using a different pair of structures, all proofs from Chapter 2 carry over to the new pair.

Definition 9. Let $i \in \{0, \omega\}$, $\beta < \omega_1^{CK}$, and $n \in \omega$. Then $\mathcal{G}_i^{\beta, n}$ is a reduced abelian p -group of length $\omega \cdot (\beta + 1)$ such that

$$U_\alpha(\mathcal{G}_i^{\beta, n}) = \begin{cases} i & \text{if } \alpha = \omega \cdot \beta + 2n \\ \omega & \text{if } \alpha = \omega \cdot \gamma + k \text{ for } \gamma \leq \beta \text{ and odd } k \\ 0 & \text{if } \alpha = \omega \cdot \gamma + k \text{ for } \gamma \leq \beta \text{ and even } k \neq 2n \end{cases}$$

Definition 10. Let $\mathcal{G}_i^{\beta, n}$ be the groups defined above, $i, j \in \{0, \omega\}$, $n \in \omega$ and $\delta \leq \beta$. Then the abstract back-and-forth relations, $\leq_\alpha$ on these groups are defined as follows.

1. $(\mathcal{G}_i^{\beta, n}, \bar{a}) \leq_{2 \cdot \delta} (\mathcal{G}_j^{\beta, n}, \bar{b})$ if and only if

- (a) the function f taking $\bar{a} = (a_0, \dots, a_n)$ to $\bar{b} = (b_0, \dots, b_n)$ by $f(a_k) = b_k$ extends to an isomorphism $\hat{f} : \langle \bar{a} \rangle \rightarrow \langle \bar{b} \rangle$,
- (b) for all $b \in \bar{b}$ and $a = f(b)$, either $h(a) = h(b) < \omega \cdot \delta$, or $h(a), h(b) \geq \omega \cdot \delta$,
- (c) for all $\gamma \leq \omega \cdot \delta$, $U_\gamma(\mathcal{G}_i^{\beta, n}) = U_\gamma(\mathcal{G}_j^{\beta, n})$.

2. $(\mathcal{G}_i^{\beta, n}, \bar{a}) \leq_{2 \cdot \delta + 1} (\mathcal{G}_j^{\beta, n}, \bar{b})$ if and only if

- (a) the function f taking $\bar{a} = (a_0, \dots, a_n)$ to $\bar{b} = (b_0, \dots, b_n)$ by $f(a_k) = b_k$ extends to an isomorphism $\hat{f} : \langle \bar{a} \rangle \rightarrow \langle \bar{b} \rangle$,
- (b) for all $b \in \bar{b}$ and $a = f(b)$, either $h(a) = h(b) < \omega \cdot \delta$ or $h(b) \geq \omega \cdot \delta$ and $h(a) \geq \min\{h(b), \omega \cdot \delta + \omega\}$,
- (c) for all $\gamma \leq \omega \cdot \delta$, $U_\gamma(\mathcal{G}_i^{\beta, n}) = U_\gamma(\mathcal{G}_j^{\beta, n})$.

Now we again verify that the hypotheses of Theorem 2.2.2 are true. We state the needed lemmas, but omit the proofs, due to their similarity to the analogous proofs from Chapter 2.

Lemma 3.2.4. The abstract back-and-forth relations in Definition 10 satisfy the back-and-forth property. Namely, let $(\mathcal{G}_i^{\beta, n}, \bar{a}) \leq_\alpha (\mathcal{G}_j^{\beta, n}, \bar{b})$ (in the sense of Definition

10). Then for any $\lambda < \alpha$ and any $\bar{d} \in \mathcal{G}_j^{\beta,n}$, there is a $\bar{c} \in \mathcal{G}_i^{\beta,n}$ such that

$$(\mathcal{G}_j^{\beta,n}, \bar{b}, \bar{d}) \leq_\lambda (\mathcal{G}_i^{\beta,n}, \bar{a}, \bar{c}).$$

Lemma 3.2.5. *Let $\mathcal{G}_0^{\beta,n}$ and $\mathcal{G}_\omega^{\beta,n}$ be as above. Then $\{G_0^{\beta,n}, \mathcal{G}_\omega^{\beta,n}\}$ is a $2(\beta+1)$ -friendly family.*

Lemma 3.2.6. *Let $\mathcal{G}_0^{\beta,n}$ and $\mathcal{G}_\omega^{\beta,n}$ be as above. Then $\mathcal{G}_\omega^{\beta,n} \leq_{2(\beta+1)} \mathcal{G}_0^{\beta,n}$.*

With these lemmas, we can now show how to extend our agreeable family of operators to successor levels.

Proposition 3.2.7. *Suppose that $(\Phi_\alpha)_{\alpha \leq \omega \cdot \beta + 2n}$ (for some fixed computable β and $n \in \omega$) is an agreeable family of Turing computable operators such that each Φ_α witnesses that $(K, \sim_\alpha^c) \leq_{tc} (ApG_{g(\alpha)}, \sim_\alpha^c)$, where*

$$g(\omega \cdot \beta + n) = \omega^2 \cdot \beta + \omega \cdot n$$

Fix an enumeration $(\psi_{\langle \gamma, j \rangle})$ of the computable infinitary sentences in the language of K such that each $\psi_{\langle \gamma, j \rangle}$ is the j th Σ_γ^c sentence in a fixed enumeration. Suppose further that for each $\rho \leq \beta$, each $\Phi_{\omega \cdot \rho + 2p}$ codes by

$$U_\eta(\Phi_{\omega \cdot \rho + 2p}(\mathcal{A})) = \begin{cases} \omega & \text{if } \eta = \omega^2 \cdot \rho + \omega \cdot m + 2j \text{ for } m < p \text{ and } \mathcal{A} \models \psi_{\langle \omega \cdot \beta + 2m, j \rangle} \\ 0 & \text{if } \eta = \omega^2 \cdot \rho + \omega \cdot m + 2j \text{ for } m < p \text{ and } \mathcal{A} \not\models \psi_{\langle \omega \cdot \beta + 2m, j \rangle} \\ \omega & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j \text{ for } \delta < \rho \text{ and } \mathcal{A} \models \psi_{\langle \omega \cdot \delta + 2m, j \rangle} \\ 0 & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j \text{ for } \delta < \rho \text{ and } \mathcal{A} \not\models \psi_{\langle \omega \cdot \delta + 2m, j \rangle} \\ \omega & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j + 1 < \omega^2 \cdot \rho + \omega \cdot p \\ 0 & \text{otherwise.} \end{cases}$$

Then there is a Turing computable operator, $\Phi_{\omega \cdot \beta + 2n+2}$ such that $(\Phi_\alpha)_{\alpha \leq \omega \cdot \beta + 2n+2}$ is

an agreeable family, and $\Phi_{\omega \cdot \beta + 2n+2}$ witnesses that

$$(K, \sim_{\omega \cdot \beta + 2n+2}^c) \leq_{tc} (ApG_{g(\omega \cdot \beta + 2n+2)}, \sim_{\omega \cdot \beta + 2n+2}^c).$$

Proof. We need to construct an operator $\Phi_{\omega \cdot \beta + 2n+2}$ that agrees with those below it, and witnesses that

$$(K, \sim_{\omega \cdot \beta + 2n+2}^c) \leq_{tc} (ApG_{g(\omega \cdot \beta + 2n+2)}, \sim_{\omega \cdot \beta + 2n+2}^c).$$

In order to do this, we first construct an intermediate operator $\hat{\Phi}$ in a way analogous to the construction in the proof of Proposition 2.2.8. Fix a computable enumeration $(\psi_i)_{i \in \omega}$ of all $\Sigma_{\omega \cdot \beta + 2n+2}^c$ sentences in the language of K . Note that determining whether or not an input structure $\mathcal{A}$ satisfies some ψ_i is $\Sigma_{\omega \cdot \beta + 2n+2}(\mathcal{A})$. Based on this, and the above back-and-forth calculations, Theorem 2.2.2 gives us a way to build a sequence of groups $(\mathcal{G}^{(i)})_{i \in \omega}$ such that

$$\mathcal{G}^{(i)} \cong \begin{cases} \mathcal{G}_{\omega}^{\omega \cdot \beta + n, i} & \text{if } \mathcal{A} \models \psi_i \\ \mathcal{G}_0^{\omega \cdot \beta + n, i} & \text{if } \mathcal{A} \not\models \psi_i. \end{cases}$$

Then, using Lemma 2.2.6, we define

$$\hat{\Phi}(\mathcal{A}) = \oplus_{i \in \omega} \mathcal{G}^{(i)},$$

a group with Ulm sequence coding the collection of $\Sigma_{\omega \cdot \beta + 2n+2}^c$ sentences true in $\mathcal{A}$ by

$$U_{\eta}(\hat{\Phi}(\mathcal{A})) = \begin{cases} 0 & \text{if } \eta = \omega^2 \cdot \beta + \omega \cdot n + 2i \text{ and } \mathcal{A} \not\models \psi_i \\ \omega & \text{if } \eta = \omega^2 \cdot \beta + \omega \cdot n + 2i \text{ and } \mathcal{A} \models \psi_i \\ \omega & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j + 1 \text{ for } m, j \in \omega, \delta < \beta \\ 0 & \text{if } \eta = \omega^2 \cdot \delta + \omega \cdot m + 2j \text{ for } m, j \in \omega, \delta < \beta. \end{cases}$$

Then, $\hat{\Phi}$ witnesses the desired embedding. We now must ensure agreement. As before, we use the direct sum operator, and define

$$\Phi_{\omega \cdot \beta + 2n+2}(\mathcal{A}) = \hat{\Phi}(\mathcal{A}) \oplus \bigoplus_{\alpha \leq \omega \cdot \beta + 2n} \Phi_{\alpha}(\mathcal{A}).$$

This is our desired operator. □

3.3 Applying Transfinite Induction on Ordinal Notation

Now that we have refined our operators so that they agree, we can obtain a result illustrating the high uniformity of this family of embeddings.

Lemma 3.3.1. *Suppose that $\alpha > 0$ is a computable ordinal of the form $\omega \cdot \beta + n$ and let the function g be defined as*

$$g(\omega \cdot \beta + n) = \begin{cases} \omega^2 \cdot \beta + \omega \cdot \frac{n}{2} & \text{if } n \text{ is even} \\ \omega^2 \cdot \beta + \omega \cdot \frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases}$$

Then, for any K , $(K, \sim_{\alpha}^c) \leq_{tc} (ApG_{g(\alpha)}, \sim_{\alpha}^c)$.

Proof. If we were to focus only on even n , Lemmas 3.2.1, 3.2.7, and 3.2.3 inductively prove this theorem, as they account for the base case, successor case, and limit case. We can generalize to odd n by noting that if K is universal for $\sim_{\alpha}^c$, then it is also universal for all $\sim_{\beta}^c$ for all $\beta < \alpha$. In the case where n is odd though, $\sim_{\alpha}^c$ no longer is equivalent to $\cong$ on $ApG_{g(\alpha)}$, which is mildly unsatisfying. □

Based on the lemma, we have a definition for a partial computable function on ordinal notations that simulates the function g . Note that each $n \in \omega$ has a unique notation, which we denote $\langle n \rangle_{\mathcal{O}}$ in the statement of Lemma 3.3.2 below.

Lemma 3.3.2. *Let a be a fixed notation for ω . Then there is a partial computable function $\hat{g} : \omega^2 \rightarrow \omega$ such that, for all $b \in \mathcal{O}$, $n \in \omega$,*

$$\hat{g}(a \cdot_{\mathcal{O}} b +_{\mathcal{O}} \langle n \rangle_{\mathcal{O}}) = \begin{cases} (a \cdot_{\mathcal{O}} a) \cdot_{\mathcal{O}} b +_{\mathcal{O}} a \cdot_{\mathcal{O}} \langle \frac{n}{2} \rangle_{\mathcal{O}} & \text{if } n \text{ is even} \\ (a \cdot_{\mathcal{O}} a) \cdot_{\mathcal{O}} b +_{\mathcal{O}} a \cdot_{\mathcal{O}} \langle \frac{n+1}{2} \rangle_{\mathcal{O}} & \text{if } n \text{ is odd.} \end{cases}$$

Proof. By lemmas 3.1.1, and 3.1.3, there are partial computable functions $+_{\mathcal{O}}$ and $\cdot_{\mathcal{O}}$. Also, given an $n \in \omega$, computing $\langle n \rangle_{\mathcal{O}}$ is a computable process, as are division and testing whether n is even. Since $\hat{g}$ is a composition of partial computable functions, $\hat{g}$ itself is partial computable. $\square$

Now, we have a way to compute the length needed for the output groups given a notation for the complexity of the input sentences, and we have a way to obtain an operator at any level.

Theorem 3.3.3. *There is a partial computable function $f : \omega \rightarrow \omega^2$, such that if $a \in \mathcal{O}$, then $f(a) = \langle e, b \rangle$, where φ_e is the operator witnessing that $(K, \sim_{|a|_{\mathcal{O}}}^c) \leq_{tc} (ApG_{|b|_{\mathcal{O}}}, \sim_{|a|_{\mathcal{O}}}^c)$.*

Proof. We proceed by transfinite induction on ordinal notation, but with base case being $\sim_2^c$ which corresponds to f receiving an input of 4.

1. Given input 4, $f(4) = \langle e, b \rangle$, where $b = \hat{g}(4) = \langle \omega \rangle_{\mathcal{O}} = a$, and e is any index for an operator such as Φ_2 in Lemma 3.2.1 which codes as desired.
2. Given input 2^c for some c , we must describe how $f(2^c)$ is computed from $f(c) = \langle e', b' \rangle$. First f uses $\hat{g}$ to compute b , the appropriate length of the output group. Then given *any* index for an operator Ψ that agrees with $\Phi_{|c|_{\mathcal{O}}}$, we must find an index for a new operator that agrees with Φ_{2^c} . Note that given any index for such a Ψ , Proposition 3.2.7 gives us a way to compute an algorithm which codes the next level of Ulm invariants, and thus a way to compute the index e for the operator $\Phi = \varphi_e$ that agrees with Φ_{2^c} . We then define $f(2^c) = \langle e, b \rangle$, as described.
3. Given input $3 \cdot 5^e$, we must describe how $f(3 \cdot 5^e)$ is computed from the set $\{f(\varphi_e(n)) : n \in \omega\}$. Let each $f(\varphi_e(n)) = \langle e_n, b_n \rangle$. Then, since all $\Phi_\alpha = \varphi_{e_n}$

agree, f outputs e described in Proposition 3.2.3, by just applying the computable direct sum operator. The length of the output group is then computed by $\hat{g}(3 \cdot 5^e) = b$. We then define $f(3 \cdot 5^e) = \langle e, b \rangle$.

□

This theorem truly highlights the uniformity of this collection of embeddings. Given this uniformity it would be reasonable to conjecture that $ApG_{\omega_1^{CK}}$ is universal for $\sim_{\omega_1^{CK}}^c$. We turn our attention to this conjecture in the next chapter. It is also pleasing that every embedding constructed in this chapter, is *tight* in the sense that ApG_β is universal for $\sim_\alpha^c$ where $\sim_\alpha^c$ is equivalent to isomorphism on ApG_β . On the other hand, this tightness only occurs for even α . It is natural to ask whether there are classes of structures K_α that are universal for $\sim_\alpha^c$, for odd α , where $\sim_\alpha^c$ is equivalent to isomorphism on K_α . If there is a large class K , that can be divided into subclasses K_α , as we did with abelian p -groups, then we could again attempt to build operators that do not interfere with one another, and lace them together with recursive transfinite induction on ordinal notation. Work in [12] shows a potential starting point.

Definition 11. Let $(p_n)_{n \in \omega}$ be the sequence of prime numbers, in ascending order. For each prime p , denote by $\mathbb{Q}_p$ the subgroup of $(\mathbb{Q}, +)$ generated by the numbers $1/p^k$ for $k \in \omega$. If $S \subseteq \omega$ is nonempty, denote by $\mathcal{G}_S$ the group

$$\mathcal{G}_S = \bigoplus_{n \in S} \mathbb{Q}_{p_n}.$$

This group is called the S -divisible torsion-free abelian group.

In [12], it is shown that the S -divisible torsion-free abelian group has a computable copy if and only if S is Σ_3^0 . The proof is by priority construction, and can be translated into a Turing computable embedding.

Theorem 3.3.4 (Downey, Goncharov, Kach, Knight, Kudinov, Melnikov, Turetsky [12]). *Let $S-DIV$ be the class of S -divisible torsion-free abelian groups. Then for any class K ,*

$$(K, \sim_3^c) \leq_{tc} (S-DIV, \sim_3^c).$$

It is clear that there is, in general, a relation between theorems of the style proven in [12], that a structure from a fixed class has a computable copy if and only if it codes a set of a certain complexity, and showing that certain classes are universal for $\sim_\alpha^c$.

CHAPTER 4

COMPACTNESS, SATURATION AND OPERATORS

As we saw in the last chapter, there is a collection of very uniform embeddings $(\Phi_\alpha)_{\alpha < \omega_1^{CK}}$ such that each Φ_α witnesses that for any class of structures K ,

$$(K, \sim_\alpha^c) \leq_{tc} (ApG_{g(\alpha)}, \sim_\alpha^c),$$

where g is the function discussed in Lemma 3.3.1. This high uniformity may lead one to speculate that ApG is universal for $\sim_{\omega_1^{CK}}^c$. Constructing an embedding witnessing such a claim has proven difficult. One approach could be to use Theorem 3.3.3 to create an operator that calls Φ_α for all computable α . This is not possible however, as there is no computable sequence of computable ordinals with limit ω_1^{CK} (since the existence of such a sequence would imply that ω_1^{CK} is computable). An approach similar in spirit that would work for some fixed computable ordinal would be to use Ash's α -system machinery, developed in [2], [3], and [4] in order “lace together” the operators $(\Phi_\alpha)_{\alpha < \beta}$. We approach the problem with Barwise-Kreisel compactness (a version of compactness for infinitary logic), as it sometimes provides a sort of limit of the Ash machinery.

It may be surprising that a compactness theorem is able to prove the existence of a Turing computable operator. After all, the standard examples of the usual compactness theorem for elementary first-order logic show the existence of orderings, groups, fields, and so on. In these examples, one axiomatizes the desired structure, shows that all finite subsets of the set of axioms have models, then applies compactness to

show that the entire set of axioms has a model. Our application is really not all that different. We axiomatize the operator, and then show that all subsets of a certain kind have models. Barwise-Kreisel Compactness then tells us that the entire set of axioms must have a model. We do hit a snag along the way though. It does not seem possible to axiomatize an operator that works for *all* structures, so we must restrict to X -computable structures for various X . We begin by examining the case where X is computable.

4.1 Analytic Sets and Compactness

In order to state the Barwise-Kreisel Compactness Theorem, we review the *analytical hierarchy* following [5]. In constructing the arithmetical hierarchy, we see that a relation $R(\bar{x})$ on natural numbers is *arithmetical* precisely when it has a definition of the form

$$R(\bar{x}) \leftrightarrow Q_1 y_1, Q_2 y_2 \dots Q_n y_n S(y_1, \dots, y_n, \bar{x}),$$

where S is a computable predicate, and the Q_i s are the number quantifiers $\forall y_i$ and $\exists y_i$. We wish to extend this notion to include relations on numbers *and functions*. In so doing, we will use f_i , $i \in \omega$ for function variables. By convention, these function variables range over total functions from ω to ω . This is done so that we may identify functions with their graphs, and perform searches that are guaranteed to converge while the graph of a function is the oracle of a Turing machine.

Definition 12. A relation $R(x_1, \dots, x_n; f_1, \dots, f_m)$ on numbers and functions, is computable if there is some e such that for all appropriate tuples $(x_1, \dots, x_n; f_1, \dots, f_m)$,

$$\varphi_e^{f_1, \dots, f_m}(x_1, \dots, x_n) = \begin{cases} 1 & \text{if } R(x_1, \dots, x_n; f_1, \dots, f_m) \\ 0 & \text{otherwise.} \end{cases}$$

As when we define the arithmetical hierarchy, we start with computable relations

and close under projection and complementation to define the analytical hierarchy. We again have a syntactic definition for our class of relations. It is not hard to show that given some relation R of the form

$$\forall x \exists f S(x, \dots; f, \dots),$$

there is an equivalent relation R' of the form

$$\exists f \forall x S'(x, \dots; f, \dots).$$

This allows us to bring all number quantifiers within the scope of all function quantifiers and arrive at the following classification.

Definition 13.

1. A relation on numbers and function $S(\bar{x}, g)$ is Σ_n^1 if it has a definition of the form

$$\exists f_1 \forall f_2 \dots Q_n f_n Q_{n+1} x R(x, \bar{x}; f_1, \dots, f_n, g),$$

where Q_n is existential if n is odd and universal if n is even, and Q_{n+1} is universal if n is odd, and existential if n is even, and R is a computable relation on functions and numbers.

2. A relation on functions and numbers is Π_n^1 if its complement is Σ_n^1 .
3. A relation on functions and numbers is Δ_n^1 if it is both Σ_n^1 and Π_n^1 .

For a synopsis of results (due to Kleene, Spector, and others) describing the characteristics of the analytical hierarchy, see [5]. Results of interest include the Π_1^1 -completeness of $\mathcal{O}$ and a result saying that a set is Δ_1^1 if and only if it is hyperarithmetical. An important relation between $\mathcal{O}$ and low levels of the analytical hierarchy is given by the following proposition.

Proposition 4.1.1. *Let f be a partial computable function defined on all of $\mathcal{O}$,*

$$\Gamma = \text{ran}(f \upharpoonright \mathcal{O}) \text{ and } \Gamma_\beta = \{f(y) : y \in \mathcal{O} \text{ and } |y|_{\mathcal{O}} < \beta\}.$$

Then Γ is Π_1^1 , and for any Δ_1^1 set $\Lambda \subseteq \Gamma$, there is a computable ordinal β such that $\Lambda \subseteq \Gamma_\beta$.

The fact that there is a computable β such that $\Lambda \subseteq \Gamma_\beta$ is often referred to as *Kleene bounding*.

We can now state the Barwise-Kreisel Compactness Theorem.

Theorem 4.1.2 (Barwise[7], Kreisel[31]). *Let Γ be a Π_1^1 set of computable infinitary sentences in some fixed computable language. Suppose that every Δ_1^1 subset of Γ is consistent. Then Γ is consistent.*

Barwise originally proved this in the context of admissible sets, while Kreisel proved a version for ω -logic. We will further discuss Barwise's context in Chapter Five.

4.2 Applying Compactness

Our goal now is to axiomatize an operator $\Phi = \phi_e$ that acts as the limit of all Φ_α constructed in Chapter 3. It is natural to ask in what language we are going to work, as the computable infinitary sentences used must *talk* about p -groups, computable functions, and computable ordinals. We formally work in the language of the arithmetic. Here, all computable functions, computable p -groups, and computable ordinals are coded via the usual Gödel coding scheme. Thus, we will informally use notation such as φ_e in the axioms below, but this will correspond to a formal expression in the language of arithmetic. There does seem to be a limitation to this approach however. We are only able to axiomatize the operator for X -computable structures for various X . We will begin by considering computable X , and will in Chapter 5 reflect on other possible X .

A good place to start is by noting that each Φ_α from Chapter 3 yields a computable function taking indices for computable structures in K to indices for computable reduced abelian p -groups. Recall that $I(K)$ is the set of indices for computable

structures in K and that the structure computed by φ_e , if any, is denoted $\mathcal{A}_e$. In the special case where $K \subseteq \text{Ap}G$, we write $\mathcal{G}_e$ for the group computed by φ_e , if any.

Observation 1. *From each $\Phi_\alpha = \varphi_e$, mapping K to $\text{Ap}G_{g(\alpha)}$, we can obtain a partial computable function f from $I(K)$ to $I(\text{Ap}G_{g(\alpha)})$ such that for all $i, j \in \omega$,*

$$f(i) = j \text{ if and only if } \Phi_\alpha(\mathcal{A}_i) = \mathcal{G}_j.$$

Proof. Given input index i , f calls Φ_α and runs φ_i to computably answer oracle questions that Φ_α asks about $\mathcal{A}_i$. Thus,

$$\Phi_\alpha(\mathcal{A}_i) = \varphi_e^{D(\mathcal{A}_i)} = \varphi_e^{\{\varphi_i(n): n \in \omega\}} = \mathcal{A}_j.$$

We have defined j as desired, and the process used to obtain it is $f = \varphi_{e'}$. □

We now apply Barwise-Kreisel Compactness. In the statement of the theorem below, we identify computable structures with their indices, and write $\text{Mod}(L)$ for the class of L -structures.

Theorem 4.2.1. *For any class K ,*

$$(I(K), \sim_{\omega_1^{CK}}^c) \leq_{tc} (I(\text{Ap}G_{\omega_1^{CK}}), \sim_{\omega_1^{CK}}^c).$$

Proof. Again, we work in the language of arithmetic, with an extra constant symbol, e . Fix computable enumerations $(\psi_{\langle \gamma, j \rangle})$ of all Σ_γ^c sentences in the language of K . As before, we identify γ with some fixed notation. This cannot be accomplished in a uniformly computable way, but it is uniformly computable in any notation for γ . We define sets Γ_α of computable infinitary formulas saying of a natural number e ,

“ e is the index for a partial computable function from
 $I(K)$ to $I(\text{Ap}G)$ that codes consistently with Φ_α .”

Thus, if we wanted to say that φ_e coded correctly the Σ_γ^c sentences true in *any* L -structure, we would want to put sentences $\theta_\gamma^3(e)$ saying

$$\forall \mathcal{A} \in \text{Mod}(L), j[(\mathcal{A} \models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A})) = \omega) \wedge (\mathcal{A} \not\models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A})) = 0)]$$

into Γ_γ . The problem here is that we have no way of *talking about* $\mathcal{A}$, and thus no way to represent the above property of φ_e as a computable infinitary sentence in the language of arithmetic. This is precisely why we have restricted ourselves to computable $\mathcal{A}$. Thus, we may write the desired statement above as

$$\theta_\gamma^3(e) \text{ if and only if } \forall i, j[(\mathcal{A}_i \in K) \rightarrow (\mathcal{A}_i \models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A}_i)) = \omega)].$$

Now, note that the expression “ $\mathcal{A}_i \models \psi_{\langle \gamma, j \rangle}$ ” can be expressed with a Σ_γ^c sentence, since $\psi_{\langle \gamma, j \rangle}$ is Σ_γ^c , and $\mathcal{A}_i$ is computable. Note that we do not even have to check whether or not the input structure is in K , as we need only check that it really is an L -structure. This is because the operators we have built work uniformly across classes K of input L -structures. Finally, as noted before, there are computable infinitary sentences saying that $U_n(\mathcal{G}) \geq m$. Thus, we can express $U_{2j}(\varphi(\mathcal{A}_i)) = \omega$ with a computable infinitary sentence. Using similar lines of reasoning, we can show that each of the following statements is definable by a computable infinitary sentence in the language of arithmetic.

1. $\theta^1(e)$ says that e is the index of a total partial computable function φ_e .
2. $\theta^2(e)$ says that given any input $\mathcal{A}_i$, $\varphi_e^{\mathcal{A}_i} \in I(\text{ApG})$.
- 3 $_\alpha$. $\theta_\gamma^3(e)$ says that $\forall i, j$ if $i \in I(\text{Mod}(L))$ then

$$[(\mathcal{A}_i \models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A}_i)) = \omega) \wedge (\mathcal{A}_i \not\models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A}_i)) = 0)].$$

- 4 $_\alpha$. $\theta_\gamma^5(e)$ says that $\forall i, j[(i \in I(\text{Mod}(L))) \rightarrow (U_{\omega \cdot \gamma + 2j+1}(\varphi_e(\mathcal{A}_i)) = \omega)]$.

Then, $\Gamma_\alpha = \{\theta^1(e), \theta^2(e)\} \cup \{\theta_\beta^k(e) : \beta \leq \alpha, k = 3, 4\}$ consists of these statements

about the number e , and the computable function φ_e , forcing it to agree with Φ_α on $I(K)$ (since it agrees on all of $\text{Mod}(L)$) up to β for all $\beta < \alpha$.

Now let $\Gamma = \cup_{\alpha < \omega_1^{CK}} \Gamma_\alpha$. By Proposition 4.1.1, this is a Π_1^1 set. We wish to use the Barwise-Kreisel Compactness Theorem to show that Γ has a model. This model would have to agree with Φ_α on computable structures for all $\alpha < \omega_1^{CK}$. This would then code the truth or falsity of *all* computable infinitary sentences into the output structure, and thus witness the universality of ApG for $\sim_{\omega_1^{CK}}^c$ amongst classes of computable structures.

We must show that the hypotheses of the Barwise-Kreisel Compactness Theorem are satisfied. Let Γ' be an arbitrary Δ_1^1 subset of Γ . By Kleene Bounding, the ordinals mentioned in the sentences in Γ have a computable least upper bound, β . Thus $\Gamma \subseteq \Gamma_\beta$. As shown by the lemmas of Chapter 3, combined with the technique of Observation 1, Γ_β has a model, the e' as in Observation 1 such that $\varphi_{e'}$ matches Φ_β on computable structures. Thus, Barwise-Kreisel Compactness applies, showing that there is an index e satisfying all of Γ . Since this index exists, we can reverse the method of Observation 1 to construct a Turing computable operator witnessing the embedding. Namely, suppose that $\varphi_e(\mathcal{A}_i) = \mathcal{G}_j$. Then we define $\Phi_{\omega_1^{CK}}(\mathcal{A}_i) = \mathcal{G}_j$, which is possible since $\mathcal{G}_j$ is computable. This $\Phi_{\omega_1^{CK}}$ is the operator witnessing the embedding. Finally, given an input structure $\mathcal{A}_i$, we know that the output group $\mathcal{G}_j = \Phi_{\omega_1^{CK}}(\mathcal{A}_i)$ will have length ω_1^{CK} . This is because we assumed that $\mathcal{A}_i$ was computable, which forces $\Phi_{\omega_1^{CK}}(\mathcal{A}_i)$ to also be computable, and Barwise and Moschovakis showed in [8] that computable reduced abelian p -groups cannot have length greater than ω_1^{CK} . $\square$

In examining the proof of Theorem 4.2.1, one may notice that the only structural properties of an input structure that are verified are that $\mathcal{A}_i \in K$, and whether or not $\mathcal{A}_i \models \psi_{\langle \gamma, j \rangle}$. This motivates the following corollary.

Corollary 4.2.2. *For any fixed computable language L , let $I(\text{Mod}(L))$ denote the*

class of (indices of) computable L -structures. Then

$$(I(\text{Mod}(L)), \sim_{\omega_1^{CK}}^c) \leq_{tc} (I(\text{ApG}_{\omega_1^{CK}}), \sim_{\omega_1^{CK}}^c).$$

Proof. We proceed as in the proof of Theorem 4.2.1, except we never need to check whether or not a computable structure $\mathcal{A}_i$ satisfies any properties other than *actually being* being a structure (checking that φ_i is total and 0-1-valued, and has value 0 for numbers that do not code sentences of the proper form). $\square$

4.3 Hyperarithmetical Saturation

In the proof of Theorem 4.2.1, the Gödel coding of all computable functions, structures, etc. depended on the presence of the natural numbers, and the operations of arithmetic, nothing more. Noting this fact, we now work in models of set theory in which ω is present. Recall that a model $\mathcal{M}$ of ZFC such that the interpretation of ω in $\mathcal{M}$ is *actually* ω is called an ω -model of ZFC . The overall idea of this section is that the reason that we were unable to axiomatize an operator that works on *all* structures in the proof of Theorem 4.2.1 is that we cannot quantify over structures. To circumvent this difficulty, we restricted ourselves to only considering computable structures. If we work in the language of set theory however, we *can* quantify over structures, as they are simply sets with some special properties. Many thanks to Robert Lubarsky for mentioning such an insight to Julia Knight and me. Fix some language, L , we define a corresponding collection of formulas $\theta_\alpha(x)$, each saying that, given an arbitrary L -structure, ϕ_x codes in a way that is agreeable with Φ_α .

Definition 14. Let $(\psi_{\langle \gamma, i \rangle})_{\gamma \leq \alpha, i \in \omega}$ be a fixed computable enumeration of L -sentences, where each $\psi_{\langle \gamma, i \rangle}$ is the i -th element of a fixed computable enumeration of all Σ_γ^c

sentences. The formula $\theta_\alpha(x)$ says that

$$(\forall \gamma \leq \alpha)(\forall i \in \omega)(\forall \mathcal{A} \in \text{Mod}(L))[(\mathcal{A} \models \psi_{\langle \gamma, i \rangle} \rightarrow U_{\omega \cdot \gamma + 2i}(\varphi_x^{D(\mathcal{A})}) = \omega)$$

$$\bigwedge$$

$$(\mathcal{A} \not\models \psi_{\langle \gamma, i \rangle} \rightarrow U_{\omega \cdot \gamma + 2i}(\varphi_x^{D(\mathcal{A})}) = 0)].$$

Again, each $\theta_\alpha(x)$ is actually a formula in the language of set theory, but since ω is definable there, so too are all the concepts used in the definition above.

Theorem 4.3.1. *If ZFC has an ω -model, $\mathcal{Z}$, then it has an ω -model in which there is an $e \in \omega$ such that for all $\alpha < \omega_1^{CK}$, $\mathcal{Z}' \models \psi_\alpha(e)$.*

Proof. We work in the language of set theory, with one additional constant symbol, e , to be interpreted as a natural number. Let Γ consist of the axioms of ZFC, along with the axiom guaranteeing an ω -model, and the *sentences* $\theta_\alpha(e)$ for all $\alpha < \omega_1^{CK}$. We use Barwise-Kreisel Compactness to show that Γ has a model. By Proposition 4.1.1, and an analysis of the axioms of ZFC, Γ is a Π_1^1 set. Choose some arbitrary Δ_1^1 set $\Gamma' \subseteq \Gamma$. We need to show that Γ' has a model. By Kleene Bounding, the ordinals mentioned in the sentences of Γ' are bounded above by some $\beta < \omega_1^{CK}$. In the real world of sets, we have shown (Theorem 3.3.3) that there is some operator Φ_β that really does work up to β on *all* input structures. Further, Φ_β exists in every ω -model of ZFC, and even has the same indices in every ω -model of ZFC. Since Φ_β works correctly on all input structures, it sure works correctly on all structures present in $\mathcal{Z}$. Thus we have that $\mathcal{Z} \models \Gamma'$, by interpreting e as any of the indices for Φ_β . Since every Δ_1^1 subset of Γ has a model, so does Γ . $\square$

The notion of *saturation* is used in model theory to formalize idea of a structure that realizes a large number of types. In particular, a countable structure $\mathcal{A}$ is *saturated* if for every tuple $\bar{a} \in \mathcal{A}$, and every set $\Gamma(\bar{a}, x)$ of formulas in the language

of $\mathcal{A}$, if every finite subset of $\Gamma(\bar{a}, x)$ has a witness in $\mathcal{A}$, then the whole set has a witness. One can think of a saturated structure as one where every element that could be forced to exist by compactness must already exist. See [33] for an excellent introduction to this area in the context of elementary first order logic. As we have seen, infinitary logic has the Barwise-Kreisel Compactness Theorem, rather than classical compactness. There is an infinitary version of saturation, due to Ressayre [35].

Definition 15. *A structure $\mathcal{A}$ is hyperarithmetically saturated if it satisfies the following conditions:*

- (a) *for all tuples $\bar{a} \in \mathcal{A}$ and all Π_1^1 sets $\Gamma(\bar{a}, x)$ of computable infinitary formulas with parameters $\bar{a}$, if every Δ_1^1 set $\Gamma'(\bar{a}, x) \subseteq \Gamma(\bar{a}, x)$ is satisfied in $\mathcal{A}$, then $\Gamma(\bar{a}, x)$ is also satisfied,*
- (b) *for all tuples $\bar{a} \in \mathcal{A}$ and all Π_1^1 sets Λ of pairs $(i, \gamma(\bar{a}))$, where $i \in \omega$ and $\gamma(\bar{a})$ is a computable infinitary sentence with parameters $\bar{a}$, if every Δ_1^1 set $\Lambda' \subseteq \Lambda$,*

$$\mathcal{A} \models \bigvee_i \bigwedge_{(i, \gamma) \in \Lambda'} \gamma(\bar{a})$$

then

$$\mathcal{A} \models \bigvee_i \bigwedge_{(i, \gamma) \in \Lambda} \gamma(\bar{a}).$$

As with saturation in elementary first order logic, we may think of a hyperarithmetically saturated structure as containing all elements that could be forced to exist via Barwise-Kreisel compactness. We now shift our attention toward obtaining a result similar in spirit to Theorem 4.2.1 using hyperarithmetical saturation.

Hyperarithmetical saturation was first studied by Ressayre in [35] in the more general context of Σ -saturation. We will further discuss this more general setting in Chapter 5. In [35], Ressayre proved a number of useful theorems on hyperarithmetical-saturation, several of which we use toward our desired result. As with saturated structures in the context of elementary first order logic, there is a notion of expansion. The second clause in the definition of hyperarithmetical saturation exists solely

to ensure that the analogy of expansion carries over to the infinitary context.

Theorem 4.3.2 (Ressayre, [35]). *Suppose that $\mathcal{A}$ is a hyperarithmetically saturated structure, and Γ is a Π_1^1 set of sentences involving symbols from the language of $\mathcal{A}$, along with finitely many new symbols. If the consequences of Γ in the language of $\mathcal{A}$ are all true in $\mathcal{A}$, then $\mathcal{A}$ can be expanded to a model $\mathcal{A}'$ of Γ . Moreover $\mathcal{A}'$ can be taken to be hyperarithmetically saturated.*

The statement of Ressayre's theorem above relies on the *consequences* of Γ in the language of $\mathcal{A}$, despite the fact that we have not discussed any formal proof system. This is fine, and we define the *consequences* of Γ in the language of $\mathcal{A}$ semantically as the set of sentences in the language of $\mathcal{A}$ true in all models of Γ .

So, we would like to apply this theorem to a hyperarithmetically saturated model of ZFC , with a new symbol representing an operator, axiomatized by a Π_1^1 set of sentences in the language of set theory. To even begin in this direction, we must first ask whether or not ZFC has a hyperarithmetically saturated model.

Theorem 4.3.3 (Ressayre, [35]). *Every consistent Π_1^1 set of computable infinitary sentences has a hyperarithmetically saturated model.*

Since ZFC is computably axiomatizable, this theorem implies a result that we want.

Corollary 4.3.4. *If ZFC has an ω -model, then it has a hyperarithmetically saturated ω -model.*

The existence of a hyperarithmetically saturated model of ZFC shows that the following result is of interest. In the theorem below, $\theta_\alpha(x)$ is as in Definition 14.

Theorem 4.3.5. *If $\mathcal{Z}$ is a hyperarithmetically saturated ω -model of ZFC , then there is an $e \in \omega$ such that for all (real-world) computable ordinals α , $\mathcal{Z} \models \theta_\alpha(e)$.*

Proof. We use clause (a) of Definition 15. Let $\Gamma(x)$ be the set of formulas, technically in the language of set theory (but we identify formulas with their coded equivalent formulations in the usual way),

$$\Gamma(x) = \{x \in \omega\} \cup \{\theta_\alpha(x) : \alpha < \omega_1^{CK}\}.$$

As seen in previous proofs, by Proposition 4.1.1, this is a Π_1^1 set. Fix some Δ_1^1 set $\Gamma' \subseteq \Gamma$. By Kleene Bounding, there is some greatest computable ordinal, say β , that appears in the formulas in Γ' . Thus,

$$\Gamma'(x) \subseteq \Gamma''(x) = \{x \in \omega\} \cup \{\theta_\alpha(x) : \alpha < \beta < \omega_1^{CK}\}.$$

As we have seen before, in the real world of sets, Theorem 3.3.3 shows that $\Gamma''(x)$ must be satisfied by some $e' \in \omega$, an index for Φ_β . Further, this index exists in every ω -model of ZFC , and works for all structures present in the model. We take this e' as the witness for x , thus showing that Γ' is satisfied in $\mathcal{Z}$. Since every Δ_1^1 subset of $\Gamma(x)$ is satisfied in $\mathcal{Z}$, $\Gamma(x)$ must also be satisfied by some natural number. This is our desired e . $\square$

It is important to note that the model $\mathcal{Z}$ of Theorem 4.3.5 is quite bizarre despite being an ω -model. First, its interpretation of ω_1^{CK} is not well-founded. This is easily seen by applying clause 1 of the definition of hyperarithmetical saturation. Further there are infinitely many sets which the model believes to be computable ordinals, which also are not even well-founded. This is why it is important to note that $\mathcal{Z} \models \theta_\alpha(e)$ for all *actual* computable α . What would it even *mean* for $\mathcal{Z}' \models \theta_\alpha(e)$ for $\alpha > \omega_1^{CK}$?

As stated, the techniques used in this chapter all depend on powerful theorems (Barwise-Kreisel Compactness, hyperarithmetical saturation) that were originally de-

veloped in a much more general framework. In the next chapter we turn toward this area.

CHAPTER 5

ADMISSIBLE SETS AND OPERATORS

In the previous chapter, we used the expressive power of $L_{\omega_1\omega}^c$ to axiomatize the Turing computable operators needed to witness universality of abelian p -groups for computable infinitary equivalence (on restricted classes of structures) in two different contexts. In Theorem 4.2.1, this was done in the language of arithmetic, and depended on Barwise-Kreisel Compactness, while Theorem 4.3.1 and Theorem 4.3.5, took place in the language of set theory, with one depending on Barwise-Kreisel Compactness, and the other on hyperarithmetical saturation. In this chapter, we look at the more general framework of Barwise and Ressayre, and use the Barwise Compactness Theorem to generalize a result from Chapter Four.

5.1 Compactness and Saturation in Admissible Set Theory

An *admissible set* is a transitive model of Kripke-Platek set theory (KP). For information on the axioms of this theory, see Appendix C. For an introduction to the theory of admissible sets, and Barwise' work in this area, please refer to [27]. Below, we give an overview of what is necessary to proceed.

Definition 16. *Let A be a countable admissible set.*

1. *The admissible fragment L_A is the set of $L_{\omega_1\omega}$ formulas present in A . Namely, $L_A = A \cap L_{\omega_1\omega}$.*
2. *We write $\mathcal{A} \preceq_{L_A} \mathcal{B}$ if $\mathcal{A}$ is an L_A -elementary substructure of $\mathcal{B}$, and $\mathcal{A} \prec_{L_A} \mathcal{B}$ if $\mathcal{A}$ is a proper L_A -elementary substructure of $\mathcal{B}$.*

3. A set is Σ_1 on A precisely in the case that it is definable in $(A, \in)$ by a finitary formula with existential quantifiers in front, followed by only bounded quantifiers.
4. A set X is A -finite if $X \in \mathcal{A}$.

We can now state the Barwise Compactness Theorem in this context.

Theorem 5.1.1 (Barwise, [7]). *Let A be a countable admissible set and T a set of L_A sentences that is Σ_1 on A . If every A -finite subset of T is consistent, then T itself is also consistent.*

Note that the Barwise-Kreisel Compactness Theorem (Theorem 4.1.2) of Chapter 4 is a special case of Theorem 5.1.1, where $A = L_{\omega_1^{CK}}$ (the ω_1^{CK} -th level of the constructible hierarchy). The admissible set $L_{\omega_1^{CK}}$ is often referred to as *the least admissible set* as it is the smallest set A in which ω is A -finite. In this case, the Π_1^1 sets are precisely the sets that are Σ_1 on A , and the Δ_1^1 sets are precisely the A -finite sets.

Just as the Barwise-Kreisel Compactness Theorem has a generalization in the context of admissible set theory, so too does Ressayre's notion of hyperarithmetical saturation. In fact, all of Ressayre's work was originally carried out for arbitrary countable admissible sets.

Definition 17. *Let A be a countable admissible set. A structure $\mathcal{A}$ is Σ_A -saturated if it satisfies the following conditions:*

1. *for all tuples $\bar{a} \in \mathcal{A}$ and all Σ_1 on A sets $\Gamma(\bar{a}, x)$ of L_A formulas with parameters $\bar{a}$, if every A -finite $\Gamma'(\bar{a}, x) \subseteq \Gamma(\bar{a}, x)$ is satisfied in $\mathcal{A}$, then $\Gamma(\bar{a}, x)$ is also satisfied,*
2. *for all tuples $\bar{a} \in \mathcal{A}$, and all A -finite sets I , and all Σ_1 on A sets Γ , consisting of pairs $(i, \gamma(\bar{a}, x))$, where $i \in I$ and $\gamma(\bar{a}, x)$ is an L_A formula, if for each A -finite $\Gamma'(\bar{a}, x) \subseteq \Gamma(\bar{a}, x)$, there is an $i \in I$ so that the set of formulas*

$$\{\gamma(\bar{a}, x) : (i, \gamma(\bar{a}, x)) \in \Gamma'(\bar{a}, x)\}$$

is satisfiable in $\mathcal{A}$, then there is an $i \in I$ so that the set

$$\{\gamma(\bar{a}, x) : (i, \gamma(\bar{a}, x)) \in \Gamma(\bar{a}, x)\}$$

is also satisfiable in $\mathcal{A}$.

As with hyperarithmetical saturation, a saturated structure can be expanded to a structure interpreting a new symbol in a way specified by an accompanying axiomatization that is Σ_1 on A if every A -finite subset of axioms is satisfiable.

Theorem 5.1.2 (Ressayre, [35]). *Let A be a countable admissible set, $\mathcal{A}$ be a Σ_A -saturated L_A -structure, and Γ be a set that is Σ_1 on A and consists of L'_A sentences, where L' is an expansion of L by finitely many new symbols. If the consequences of Γ in the language of L_A are all true in $\mathcal{A}$, then $\mathcal{A}$ can be expanded to a model $\mathcal{A}'$ of Γ . Moreover $\mathcal{A}'$ can be taken to be hyperarithmetically saturated.*

5.2 Σ_1 is Necessary for Gregory's Theorem

Admissible set theory provides a unified framework for model theory of infinitary logic, and generalized computability theory. One of the many contributors to this area was John Gregory, who was a student of Carol Karp. Early work of Gregory [19] included the following theorem.

Theorem 5.2.1 (Gregory, [19]). *Let A be a countable admissible set. Suppose T is a set of L_A sentences that is Σ_1 on A . If T has a pair of countable models $\mathcal{A}$ and $\mathcal{B}$ such that $\mathcal{A} \prec_{L_A} \mathcal{B}$, then T has an uncountable model.*

Gregory claimed that there were known examples that demonstrated the necessity of T being Σ_1 on A . After finding no such examples in the literature, nor any examples known by experts in the field, Knight suggested that her research group try to construct such an example. In [23], Johnson, Knight, Ocasio, and the present author did just that.

Theorem 5.2.2 (Johnson, Knight, Ocasio, V., [23]). *There is a set T of computable infinitary sentences, in a computable language, L , such that T has (up to isomorphism) exactly two models, $\mathcal{A}$ and $\mathcal{B}$, where $\mathcal{A} \prec_{L_{\omega_1^{CK}}} \mathcal{B}$. Further, for each computable ordinal α , the set of computable Σ_α sentences in T is hyperarithmetical.*

The proof of this theorem relies on iterated forcing to actually construct the models, the underlying idea of the construction is not so difficult to explain. We give a sketch of the idea of the proof following some definitions that we need.

Let α be a computable ordinal. For any index set I , a relation $R \subseteq I \times \omega$ can be thought of as representing a collection of subsets of ω by setting the set with index i to be $\{n : (i, n) \in R\}$. Let $2^{<\omega}$ denote the set of functions from finite subsets of ω to 2, and $2^{<I \times \omega}$ denote the set of functions from finite subsets of $I \times \omega$ to 2.

Definition 18. 1. A set $X \subseteq \omega$ is α -generic if for each Σ_α^0 set $D \subseteq 2^{<\omega}$, there is some $p \subseteq \chi_X$ such that either $p \in D$ or else there is no $q \supseteq p$ such that $q \in D$.

2. Let I be a computable set of indices. An indexed family $R \subseteq I \times \omega$ is α -generic if for each Σ_α^0 set $D \subseteq 2^{<I \times \omega}$, there is some $p \subseteq \chi_R$ such that either $p \in D$ or else there is no $q \supseteq p$ such that $q \in D$.

In some sense, a set $X \subseteq \omega$ is α -generic if there is really nothing special about it, as far as $\Delta_{\alpha+1}^0$ can tell.

Proof. (Sketch of Proof for Theorem 5.2.2) We let L be the language $\{U_n : n \in \omega\}$, consisting of countably many unary predicates. In an L -structure $\mathcal{M}$, each element $m \in \mathcal{M}$ represents an associated set, $\{k \in \omega : U_k(m)\}$. We enforce, by adding sentences to T , that in models of T , distinct elements correspond to distinct sets of natural numbers. The universe of $\mathcal{A}$ is an infinite computable set of constants C , partitioned into infinitely many infinite sets C_n . The universe of $\mathcal{B}$ is the universe of $\mathcal{A}$, but with one extra constant, b . In $\mathcal{A}$, all sets represented will be hyperarithmetical, and in all models of T , any hyperarithmetical sets represented are the same as those represented in $\mathcal{A}$. In $\mathcal{B}$, there is one set represented that is not hyperarithmetical,

and this set is represented by b . Further, in any model of T , if there is a non-hyperarithmetical set represented, then we enforce that it must be the same set as that represented by b in $\mathcal{B}$. We define T as the set of computable infinitary sentences true in $\mathcal{A}$ and $\mathcal{B}$.

Now the idea is to ensure that $\mathcal{A} \prec_{L_{\omega_1^{CK}}} \mathcal{B}$ by constructing b and the $c \in C_n$ in such a way that the sequence of sets represented by C_n *approximates* the set represented by b . Namely, we will have that for all $c \in C_n$ and $k < n$, k is in the set represented by c if and only if it is in the set represented by b . In order to ensure that no computable infinitary sentences are able to *pinpoint* the difference between $\mathcal{A}$ and $\mathcal{B}$, we need to guarantee some level of genericity of our structures. In particular, the set represented by b must be α -generic for all $\alpha < \omega_1^{CK}$, and the sets represented by $c \in C_n$ are an α_n -generic family that is $\Delta_{\alpha_n+1}^0$, where $(\alpha_n)_{n \in \omega}$ is a sufficiently quickly increasing sequence of computable ordinals with limit ω_1^{CK} .

From this point, we can already show that $\mathcal{A}$ and $\mathcal{B}$ are the only models of T up to isomorphism. In order to show that $\mathcal{A} \prec_{L_{\omega_1^{CK}}} \mathcal{B}$, we need a formal way to track the truth of complicated statements in $\mathcal{A}$ and $\mathcal{B}$. This is where the forcing machinery comes in. First we construct some hyperarithmetically generic set, B to be represented by b . Then, we construct the family of sets that will *rapidly approximate* B . This family consists of $R_n \subseteq C_n \times \omega$, where each R_n is α_n -generic and $\Delta_{\alpha_n+1}^0$. These give us a way of realizing the sets that are represented by each $c \in C_n$. Then we can realize each of $\mathcal{A}$ and $\mathcal{B}$ as limit structures by letting $\mathcal{A}_n$ be the structure with universe $\cup_{k \leq n} C_k$, and $\mathcal{B}_n$ be the structure with universe $\cup_{k \leq n} C_k \cup \{b\}$. The purpose of forming these structures is to approximate truth in $\mathcal{A}$ and $\mathcal{B}$. Again, this is carried out by forcing. Next, the following facts are proven by the examining the truth and forcing lemma, combined with the iterated forcing that was used to construct the sets:

- $\mathcal{A}_n \prec_{\alpha_n} \mathcal{A}_{n+1}$,

- $A_n \prec_{\alpha_n} \mathcal{B}_n$,
- $\mathcal{B}_n \prec_{\alpha_n} \mathcal{B}_{n+1}$.

When combined, these facts show that $\mathcal{A} \prec_{L_{\omega_1^{CK}}} \mathcal{B}$, as desired. $\square$

5.3 Applying Barwise Compactness

We now turn to generalizing Theorem 4.2.1 to the context of admissible set theory. As stated above, $L_{\omega_1^{CK}}$ is often referred to as the *least admissible set*, since it is the smallest admissible set that has ω as an element. If we add to $L_{\omega_1^{CK}}$ new sets of natural numbers, none of which compute any new (i.e. non-computable) ordinals, then we obtain a *fattening of the least admissible set*. This terminology comes from picturing the levels of Gödel’s constructible universe as triangles of varying heights. If A is a countable admissible set, and $A \supset L_{\omega_1^{CK}}$ that only contains the computable ordinals, then A is “fatter” (it contains new sets), but the same “height” (no new ordinals). For $X \subseteq \omega$, let ω_1^X denote the least ordinal not computable from X . Thus, for hyperarithmetical X , $\omega_1^X = \omega_1^{CK}$. There are non-hyperarithmetical sets X such that $\omega_1^X = \omega_1^{CK}$.

Let $I^X(K)$ be the set of indices for structures in K that are computable relative to X . Further, let the structure computed by φ_e^X be denoted by $\mathcal{A}_e^X$. As before, in the special case where we are discussing groups, we let $\mathcal{G}_e^X$ denote the group computed by φ_e^X . Now, the Barwise Compactness Theorem allows us generalize Theorem 4.2.1 to show that the class of abelian p -groups is universal for $\sim_{\omega_1^{CK}}^c$ not only amongst computable structures, but for X -computable structures, for any X that lives in a fattening of the least admissible set.

As with the proof of Theorem 4.2.1, we first note that each Φ_α yields a corresponding partial computable function on indices of structures.

Observation 2. From each $\Phi_\alpha = \varphi_e$, mapping K to $ApG_{g(\alpha)}$, we can obtain a partial computable function f from $I^X(K)$ to $I^X(ApG_{g(\alpha)})$ such that for all $i, j \in \omega$,

$$f(i) = j \text{ if and only if } \Phi_\alpha(\mathcal{A}_i^X) = \mathcal{G}_j^X.$$

Proof. The proof of the observation is similar to before, but slightly more subtle. Given index $i \in I^X(K)$, f calls Φ_α , and tells it to run φ_i with oracle X to answer questions about the input structure, $\mathcal{A}_i^X$. Then we have that

$$\Phi_\alpha(\mathcal{A}_i^X) = \varphi_e^{\mathcal{A}_i^X} = \mathcal{G}_j^X.$$

We have obtained j as desired, and the process used to obtain it is $f = \varphi_{e'}$. $\square$

Note that f is *truly computable*, not merely an X -computable function. As with the relativized $s - m - n$ Theorem, this comes from the fact that f is uniform with respect to X .

Theorem 5.3.1. Let $X \subseteq \omega$ be such that $\omega_1^X = \omega_1^{CK}$. For any class K ,

$$(I^X(K), \sim_{\omega_1^{CK}}^c) \leq_{tc} (I^X(ApG), \sim_{\omega_1^{CK}}^c).$$

Proof. Fix some $X \subseteq \omega$, such that $\omega_1^X = \omega_1^{CK}$. Thus, X is an element of an admissible set A , some fattening of the least admissible set. Also fix computable enumerations $(\psi_{\langle \gamma, j \rangle})$ of all Σ_γ^c sentences in the language of K . As before, we wish to specify a set $\Gamma(e)$ of formulas about a natural number, e , such that e satisfies Γ if and only if φ_e corresponds to an operator witnessing the desired embedding. As we saw earlier, the only difficulty with this approach is *actually writing down* the formulas axiomatizing e . We first define sets Γ_α of computable infinitary formulas saying of a natural number e

“ e is the index for a partial computable function from $I^X(K)$ to $I^X(ApG)$ that codes consistently with Φ_α .”

Again, the only difficulty is writing sentences capable of *talking about* the structures in K . This is why we have restricted ourselves to X -computable structures. So, if we wished to write a sentence saying about e , that φ_e correctly (in a way that agrees with Φ_γ) codes the Σ_γ^c sentences *true* in all input structures from $I^X(K)$ into the output groups, we would write a sentence $\theta_\gamma^{3,X}$ saying that for all $i, j \in \omega$, if $i \in I^X(K)$, then

$$[(\mathcal{A}_j^X \models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A}_j^X)) = \omega) \wedge (\mathcal{A}_i^X \not\models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e(\mathcal{A}_i^X)) = 0)].$$

Note that the expression “ $\mathcal{A}_i^X \models \psi_{\langle \gamma, j \rangle}$ ” can be expressed with a sentence that is Σ_α^c *relative to X* , since $\psi_{\langle \gamma, j \rangle}$ is Σ_γ^c and $\mathcal{A}_i^X$ is computable relative to X . As in the proof of Theorem 4.2.1, we do not need to actually check that $\mathcal{A}_i^X \in K$, since our operator will work for all L -structures. Finally, the output groups will also be computable relative to X , so we can write computable infinitary sentences, relative to X , saying that the Ulm invariants have certain values. Using similar lines of reasoning, one can see that each of the following statements is definable by a computable infinitary sentence, relative to X , in the language of arithmetic.

1. $\theta^{1,X}$ says that e is the index of a total partial computable function φ_e .
2. $\theta^{2,X}$ says that given any input $\mathcal{A}_i^X$, $\varphi_e^{\mathcal{A}_i^X} \in I(ApG)$.
3. $\theta_\alpha^{3,X}$ says that $\forall i, j$ if $\mathcal{A}_i^X$ is an L -structure, then

$$[(\mathcal{A}_i^X \models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e^{\mathcal{A}_i^X}) = \omega) \wedge (\mathcal{A}_i^X \not\models \psi_{\langle \gamma, j \rangle} \rightarrow U_{\omega \cdot \gamma + 2j}(\varphi_e^{\mathcal{A}_i^X}) = 0)].$$

4. $\theta_\alpha^{5,X}$ says that $\forall i, j [(\mathcal{A}_i^X \in K) \rightarrow (U_{\omega \cdot \gamma + 2j + 1}(\varphi_e^{\mathcal{A}_i^X}) = \omega)]$.

Now, for $\alpha < \omega_1^{CK}$, define $\Gamma_\alpha = \{\theta^{1,X}, \theta^{2,X}\} \cup \{\theta_\beta^{k,X} : \beta \leq \alpha, k = 3, 4\}$. Thus, Γ_α is satisfied if and only if there is some $e \in \omega$ such that the function φ_e corresponds to an operator that agrees with Φ_α on $I^X(K)$ up to β for all $\beta < \alpha$.

Now let $\Gamma = \cup_{\alpha < \omega_1^{CK}}$. Since X computes no new ordinals, this set is Σ_1 on A . If we can show that the hypotheses of the Barwise Compactness Theorem are satisfied, then there will be some e satisfying Γ , which must then correspond to the desired operator. Let Γ' be an arbitrary A -finite subset of Γ . Since A is an admissible set, it satisfies Δ_0 -collection, showing that Γ' is an element of A , and thus has some upper bound $\beta < \omega_1^{CK}$ on the ordinals which appear in its formulas (this is analogous to Kleene bounding). The lemmas of Chapter 3, combined with 2 then show that Γ_β , and thus Γ' has a model. Thus, Barwise Compactness applies, showing that all of Γ is consistent. The image of this operator must be $I^X(ApG)$, since all output groups are computable in the input structures, all of which are X -computable. Finally, it is worth noting that the work of Barwise and Moschovakis in [8] shows that the lengths of the output groups must still be computable ordinals. $\square$

5.4 A Question on Σ_A -Saturation

It is very natural now to wish to generalize Theorem 4.3.5 to the context of admissible sets. Indeed, one might think that we can simply follow analogy to arrive at the following.

Non-Theorem 1. *Let $\alpha > \omega$ be a countable admissible ordinal (namely an ordinal such that L_α is an admissible set). If ZFC has an ω -model, then it has an ω -model $\mathcal{Z}'$, in which there is an $e \in \omega$ such that for all $\beta < \alpha$, $\mathcal{Z}' \models \theta_\beta(e)$ (amongst class of structures present in the model), where $\theta_\beta(e)$ says ‘ φ_e is an embedding witnessing that ApG is on top for $\sim_\beta^c$ ’*

I am not claiming that this is not true, just that we are unable to directly port the proof of Theorem 4.3.5 to a proof of the supposed Theorem above. The proof does not translate, due to the fact that new ordinals, $\gamma > \omega_1^{CK}$ can be elements of L_α , and we have said nothing about universality of p -groups with respect to $\sim_\gamma$ for

such a γ . It is true (see [8]) that the least admissible set is in some sense, “as thin as possible.” Namely, given its height, it has no more sets than it must in order to be admissible. In Theorem 5.3.1, we showed that we can carry out universality arguments if we fatten this set. Thus, it would be natural to have a companion result on heightening the admissible set in question. However, more hypotheses, on this notion of “thinness” would be needed. In such a situation, it may be possible to further develop Ash’s machinery in order to produce operators at these levels. In this context, the sentences discussed would no longer be *computable* infinitary, as we would be looking at a larger fragment of $L_{\omega_1\omega}$. Thus, rather than *computable infinitary equivalence*, we would define L_α -equivalence, denoted $\sim_{L_\alpha}$, and say that two structures $\mathcal{A}$ and $\mathcal{B}$ are L_α equivalent if they satisfy precisely the same sentences in this thin fragment L_α .

Conjecture 1. *Under suitable thinness hypotheses on an admissible set A , there is an operator witnessing that the class of abelian p -groups, whose Ulm sequences can be distinguished by the sentences present in L_α are universal for L_α -equivalence.*

It is possible that there are other natural generalizations of Theorem 4.3.5 to Barwise’s original context, but this seems the most natural.

APPENDIX A

ABELIAN p -GROUPS

For a beautiful exposition of the theory of infinite abelian groups, see [24]. An abelian group $\mathcal{G}$ is said to be a p -group if, for some fixed prime p , every non-zero element of $\mathcal{G}$ has order some power of p . The class of p -groups is of special interest to group theorists, since they serve as building blocks of more complicated groups.

Theorem A.0.1. *Every torsion group is a direct sum of p -groups (over varying p).*

In abelian p -groups, we may multiply any element by an integer by recognizing multiplication as iterated addition. Namely, for $x \in \mathcal{G}$ and $n \in \omega$,

$$nx = \underbrace{x + \cdots + x}_{n \text{ times}}.$$

We can extend this to the integers by letting $-nx = n(-x)$. Since there is a notion of multiplication, it is natural to wonder whether there is a meaningful way to interpret division. It is only sometimes the case that division makes sense. If there is a y such that $ny = x$, then we say that n *divides* x or that x *is divisible by* n . Many well-known groups, such as the additive group of rationals, have the property that division *always* makes sense. This property is worth defining.

Definition 19. *An element x of a group is divisible if for every $n \in \mathbb{Z}$, n divides x . A group $\mathcal{G}$ is said to be divisible if every element x of $\mathcal{G}$ is divisible.*

Note that a direct sum of groups is divisible if and only if every summand is divisible. As an example of a countable divisible abelian p -group, consider the Prüfer p -group, $Z(p^\infty)$. There are several common presentations of this group.

Definition 20. Let p be some fixed prime, and let $(P, +)$ be the additive subgroup of $(\mathbb{Q}, +)$ with universe $P = \{n/p^m : n \in \mathbb{Z}, m \in \omega\}$. The Prüfer p -group, denoted $Z(p^\infty)$ is $P/\mathbb{Z}$.

To gain some familiarity with the Prüfer p -groups, consider the case where $p = 3$. Here, the group consists of all rational numbers in the interval $[0, 1)$ of the form $m/3^n$, and addition is carried out mod one. For example

$$1/3 + 1/3 = 2/3, \quad 1/3 + 2/3 = 0, \quad 8/9 + 8/9 = 7/9, \quad 2/3 + 8/9 = 5/9,$$

and so on. Another presentation of this group is as the set of p^n -th roots of unity as a multiplicative subgroup of the circle group in the complex plane. The additive group of rationals and the Prüfer p -groups act as atomic summands in a decomposition theorem for divisible abelian p -groups.

Theorem A.0.2. A divisible abelian group is a direct sum of groups, each of which is isomorphic to the additive group of rationals, or to $Z(p^\infty)$, the Prüfer p -group (over varying p).

A divisible abelian group is a group where division of any element by any integer is well-defined. At the opposite end of the spectrum, a *reduced* abelian p -group is one with no divisible elements. We introduce the notion of *height* to aid in analyzing the degree of divisibility of an element.

Definition 21. Let $\mathcal{G}$ be a p -group. Then non-zero $x \in \mathcal{G}$ has finite height n if x is divisible by p^n but not p^{n+1} .

To extend this notion to infinite height, we will need some notation. We write $p\mathcal{G}$ for $\{pg : g \in \mathcal{G}\}$, and define $p^{\alpha+1}\mathcal{G} = p(p^\alpha\mathcal{G})$. At limit ordinals β we take intersections, letting $p^\beta\mathcal{G} = \bigcap_{\alpha < \beta} p^\alpha\mathcal{G}$ for limit β . Since we are assuming that $\mathcal{G}$ is reduced, there are no divisible elements, thus there is some ordinal λ at which this

process ends. Namely λ is the least ordinal such that $p^\lambda \mathcal{G} = p^{\lambda+1} \mathcal{G} = \{0\}$. We call this ordinal the *length* of $\mathcal{G}$, and denote it $\lambda(\mathcal{G})$.

Definition 22. For $\alpha < \omega_1$, we say that non-zero x has infinite height α if $x \in p^\alpha \mathcal{G} \setminus p^{\alpha+1} \mathcal{G}$.

To each countable reduced abelian p -group, we may associate a countably transfinite sequence of natural numbers and ω . This sequence is called the *Ulm sequence* of the group, and plays a very special role in the theory of classification of groups.

Theorem A.0.3 (Ulm). *Two countable reduced abelian p -groups are isomorphic if and only if they have the same Ulm sequences.*

The α th element of this sequence is denoted $U_\alpha(\mathcal{G})$. Now, since each $p^{\alpha+1} \mathcal{G} \triangleleft p^\alpha \mathcal{G}$, we may form the quotient, $p^\alpha \mathcal{G} / p^{\alpha+1} \mathcal{G}$, a $\mathbb{Z}_p$ vector space. In abelian p -groups, the elements of order p act as a sort of *foundation* for the group, so define the subgroup $P = \{x \in \mathcal{G} : px = 0\}$. Then, let $p^\alpha P = P \cap p^\alpha \mathcal{G}$. As before, we may form $p^\alpha P / p^{\alpha+1} P$, a $\mathbb{Z}_p$ vector space. Let $U_\alpha(\mathcal{G}) = \dim(p^\alpha P / p^{\alpha+1} P)$. This is the α -th *Ulm invariant*, and the α -th element of the *Ulm sequence* of $\mathcal{G}$.

With the machinery used in the construction of the Ulm invariants, we can now extend to notion of height to infinite ordinals.

Definition 23. Let $\mathcal{G}$ be a p -group. Then non-zero $x \in \mathcal{G}$ has height α if $x \in p^\alpha \mathcal{G} \setminus p^{\alpha+1} \mathcal{G}$.

There are two noteworthy ways to visualize abelian p -groups. First, in [37], Laurel Rogers describes a way of relating trees and p -groups. The nodes of the tree generate the group. To each x in $\mathcal{G}$, we associate a node, denoted $n(x)$. The successor relation $n(x)Sn(y)$ on nodes in the tree holds if and only if $px = y$ for $x \neq y$. She then goes on to describe a new proof of Ulm's theorem using these trees and a collection of *stripping functions*. The second visualization applies to countable reduced abelian

p -groups of length $\lambda = \omega \cdot (\beta + n)$. We often picture not the group itself, but the Ulm sequence of the group, and the visualization is really just an alternate way to picture a transfinite sequence. This is done by representing the Ulm sequence as a matrix with entries from $\omega + 1$, of height $\omega \cdot (\beta + n)$ and width ω . For example, a countable reduced abelian p -group $\mathcal{G}$ of length $\omega^2 + \omega$ would have the matrix

$$\text{height } \omega + 1 \left\{ \begin{array}{cccccc} U_{\omega^2}(\mathcal{G}) & U_{\omega^2+1}(\mathcal{G}) & U_{\omega^2+2}(\mathcal{G}) & U_{\omega^2+3}(\mathcal{G}) & U_{\omega^2+4}(\mathcal{G}) & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \\ U_{\omega \cdot 2}(\mathcal{G}) & U_{\omega \cdot 2+1}(\mathcal{G}) & U_{\omega \cdot 2+2}(\mathcal{G}) & U_{\omega \cdot 2+3}(\mathcal{G}) & U_{\omega \cdot 2+4}(\mathcal{G}) & \dots \\ U_{\omega}(\mathcal{G}) & U_{\omega+1}(\mathcal{G}) & U_{\omega+2}(\mathcal{G}) & U_{\omega+3}(\mathcal{G}) & U_{\omega+4}(\mathcal{G}) & \dots \\ U_0(\mathcal{G}) & U_1(\mathcal{G}) & U_2(\mathcal{G}) & U_3(\mathcal{G}) & U_4(\mathcal{G}) & \dots \end{array} \right.$$

where there are infinitely many $m \in \omega$ such that $U_{\omega^2} + m(\mathcal{G}) \neq 0$.

APPENDIX B

ORDINAL NOTATION

In defining the *hyperarithmetical hierarchy*, some care must be taken in how we represent Σ_α sets for $\alpha \leq \omega_1^{CK}$. In particular, we would like a way to represent each computable ordinal by a natural number in a meaningful way. In [29], Kleene defines $\langle \mathcal{O} \subset \omega, | \cdot |_O, <_O \rangle$ as a structure for indexing the computable ordinals, where $| \cdot |_O$ is a function from $\mathcal{O}$ to ω_1^{CK} and $<_O$ is a partial ordering of $\mathcal{O}$. This system is called Kleene's $\mathcal{O}$, and following his conventions, we refer to an element $a \in \mathcal{O}$ such that $|a|_O = \alpha$ as a *notation for α* .

Kleene's $\mathcal{O}$ is defined inductively. First define $|1|_O = 0$, and let $|a|_O = \alpha$ imply that $|2^a|_O = \alpha + 1$. These definitions give unique notations to all natural numbers: $|1|_O = 0$, $|2|_O = 1$, $|4|_O = 2$, and in general $|2^{2^{\dots^2}}|_O = n$, where $2^{2^{\dots^2}}$ is an exponential tower of height n . We also inductively define $a <_O 2^b$ if $a <_O b$ or $a = b$. This process extends to limit ordinals α by setting $|3 \cdot 5^e| = \alpha$ if and only if φ_e is a total computable function, with range a subset of $\mathcal{O}$, so that

$$\varphi_e(0) <_O \varphi_e(1) <_O \varphi_e(2) <_O \dots,$$

where $\alpha = \sup_{n \in \omega} |\varphi_e(n)|$. Finally, define $a <_O 3 \cdot 5^e$ if there is an n such that $a <_O \varphi_e(n)$. Note that this system gives unique notations to each finite ordinal, but infinitely many to each infinite computable ordinal. This is due to two reasons. First, padding ensures that for any notation $3 \cdot 5^e$ for a limit ordinal, there are infinitely many e' such that $\varphi_e = \varphi_{e'}$, which means that $3 \cdot 5^{e'}$ is also a notation for the same

limit ordinal. Secondly, given a computable limit ordinal α , there are infinitely many computable sequences of ordinals with limit α . This also guarantees the existence of infinitely many notations for each infinite ordinal.

Since there are only countably many notations, and uncountably many countable ordinals, there must be some least countable ordinal for which there is no notation. This ordinal turns out to be ω_1^{CK} , and indeed Spector showed that an ordinal is computable if and only if it has a notation. In proving this correspondence, Spector also demonstrated many desirable properties of $\mathcal{O}$.

Proposition B.0.4. *If $|a|_{\mathcal{O}} = \alpha$, then the set $\{b : b <_{\mathcal{O}} a\}$ consists of exactly one notation for each $\beta < \alpha$. Further, this set, under the relation of $<_{\mathcal{O}}$, is isomorphic to $\text{pred}(\alpha)$ via the isomorphism $| \cdot |_{\mathcal{O}}$.*

This proposition allows us to normally identify a computable ordinal α with some fixed notation a , then work solely with $b <_{\mathcal{O}} a$ when discussing $\beta < \alpha$. Although the relation $<_{\mathcal{O}}$ is not a computably enumerable relation on ω (indeed, $\mathcal{O}$ is actually Π_1^1), it is c.e. on *initial segments* of $\mathcal{O}$. In fact, there are Π_1^1 paths through $\mathcal{O}$ with exactly one notation for each ordinal.

Proposition B.0.5. *There is a computable function p such that for all $b \in \mathcal{O}$ there is an $e \in \omega$ such that,*

$$\{a : a <_{\mathcal{O}} b\} = W_{p(e)}.$$

This allows us to, given a notation a for α , enumerate notations for all $\beta < \alpha$ in a way that is *meaningful* relative to a .

APPENDIX C

KRIPKE-PLATEK SET THEORY

Kripke-Platek set theory, denoted KP , is a weakening of ZF . Recall that a formula in the language of set theory is Δ_0 precisely if it has no unbounded quantification (and is finitary). It consists of the following axioms.

1. Extensionality
2. Union
3. Pairing
4. Foundation
5. Δ_0 -Separation scheme: Let $\varphi(x, \bar{y})$ be a Δ_0 formula with no free occurrence of v . Then we have the axiom

$$(\forall \bar{y})(\forall v)(\exists u)(\forall x)(x \in u \leftrightarrow [x \in v \wedge \varphi(x, \bar{y})]).$$

6. Δ_0 -Collection scheme: Let $\varphi(x, y, \bar{z})$ be a Δ_0 formula with no free occurrence of v . Then we have the axiom

$$(\forall \bar{z})(\forall u)[(\forall x \in u)(\exists y)\varphi(x, y, \bar{z}) \rightarrow (\exists v)(\forall x \in u)(\exists y \in v)\varphi(x, y, \bar{z})].$$

Recall that a Σ formula is (a finitary) one that has no unbounded universal quantification. In the axioms above, items 5 and 6 give schema for Δ_0 -collection. It is useful to note that collection for Σ formulas is a theorem of KP . For more background, see [27] or [26].

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